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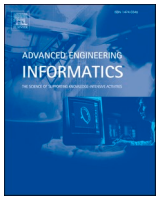

Full length article

Digital twin-driven intelligent operation and maintenance platform for large-scale hydro-steel structures

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ABSTRACT

Large-scale hydro-steel structures (LS-HSS) are pivotal in hydraulic engineering, boasting high lift, expansive apertures, and discharge capacities. Conventional methods of operation and maintenance for LS-HSS require a digital revolution to ensure safety, reliability, and durability. This study introduces a novel approach, harnessing the power of a digital twin (DT) to integrate physical and virtual elements in LS-HSS operation and maintenance seamlessly. A digital twin modeling framework of LS-HSS operation and maintenance is first presented, incorporating five levels of a physical entity, a virtual entity, DT data, services, and connections. Subsequently, a DT-driven intelligent operation and maintenance (DT-IOM) platform is constructed and applied, which features real-time mapping, closed-loop control, dynamic health assessment, and intelligent decision-making. This platform provides a pathway for digitally transforming conventional LS-HSS operation and maintenance and paves the way for a new era of intelligent infrastructure management. The proposed approach has been successfully implemented and validated with multiple engineering projects, including radial gates of reservoir spillways, plain gates of flood discharge orifices, and emergency gates of pumped storage power stations. The impressive results demonstrate a significant anomaly response and processing efficiency improvement of over 60%. Moreover, the platform has substantially reduced field inspection time and equipment failure rate by approximately 50% and 20%, respectively. These tangible benefits further highlight the platform's functional advantages, including its capability for IoT connection, modeling and integration, virtual-real interaction, 2D/3D visualization, and service expansion. The results demonstrate the unique features in the closed loop of perception, analysis, decision-making, and optimization for DT-IOM, especially for integrating data, models, knowledge, and services in LS-HSS intelligent operation and maintenance.

1. Introduction

With the rapid advancement of emerging information technologies, such as the Internet of Things (IoT) [1], Big Data [2], and Artificial Intelligence (AI) [3], there is currently active exploration in the research field to finely describe, accurately diagnose, and intelligently make decisions for operational processes of objects in the physical world by continuously enhancing digital mapping capabilities [4]. During the global transition to industrial intelligence, the digital twin (DT) is a unified entity encompassing both the physical object and its digital representation, enabling seamless interconnectivity, fostering collaboration, and co-evolving. It possesses characteristics such as the integration of virtual and real, real-time interaction, iterative optimization, and data-driven approach [5], thus attracting extensive attention.

The integration and widespread adoption of digital technology in conventional sectors have led to the progressive utilization of digital twin technology for engineering structures [6,7]. This has particularly resulted in significant implications for aging, antiquated, and hazardous structures. Therefore, in the context of extending the service life of hydraulic engineering projects and the increasingly prominent demands for safety monitoring and health management of engineering facilities and equipment [8], the integration of digital twin technology into the conventional operation and maintenance process is gradually being implemented to ensure comprehensive safety throughout the entire service life. However, relevant research in hydraulic engineering primarily focuses on areas such as engineering construction management [9–11], dam safety monitoring [8,12,13], hydroelectric generating units [14,15], and water resource management [16,17]. Insufficient attention

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1474-0346/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

has been given to the impact of digital twin technology on enhancing the performance of large-scale hydro-steel structures (LS-HSS) intelligent maintenance management and corresponding digital twin platform research. The digital-enabled operation and maintenance methods for LS-HSS primarily focus on monitoring, information modeling, multi-physics modeling, simulation fusion, or dynamic model correction (as illustrated in Table 1). The limitations of efficiency and reliability in health monitoring, assessment, and decision-making based on sensors' signals, historical data, or specific static physical /simulation data for LS-HSS underscore the necessity to transition towards cyber-physical integration within the conventional maintenance process. The current methodologies, however, do not encompass a comprehensive framework of digital twin modeling for LS-HSS operation and maintenance. Moreover, exploring integrating methods of data, models, knowledge, and services driven by digital twin technology is essential. Subsequently, establishing a comprehensive DT-driven intelligent operation and maintenance (DT-IOM) platform becomes crucial in promoting the digital transformation of conventional LS-HSS operations.

Simultaneously, LS-HSS have a profound and enduring impact on hydraulic engineering. They undertake vital functions such as flood management, water supply for diverse purposes, irrigation facilitation, power generation, and navigation. Many of them have been extended to provide additional services. However, the issue of safety remains prominent [25]. Moreover, the construction of high dams and giant hydroelectric units, coupled with the enhancement of steel structures' manufacturing capabilities, has catalyzed the advancement of hydro-steel structures with higher capacity for head, flow, and discharge [26]. The future will witness a widespread application of LS-HSS in crucial projects, such as floodgates in vast reservoirs (e.g., Xiaolangdi) [27], navigation equipment in major waterways (e.g., ship locks and ship lifts in the Three Gorges Project) [28], and giant hydroelectric power station (e.g., floodgates at Baihetan Power Station) [29]. As these structures operate under complex conditions, increasing demand exists to enhance their intelligent operation and maintenance continuously.

This study proposes a five-dimensional digital twin modeling framework of LS-HSS operation and maintenance, specifically focusing on the digital transformation pathway of conventional operation and maintenance and the integration methods of data, models, knowledge, and services driven by digital twin technology. Then, a DT-driven LS-HSS intelligent operation and maintenance (DT-IOM) platform features real-time mapping, closed-loop control, dynamic health assessment, and intelligent decision-making is constructed and applied. The DT-IOM platform establishes a closed-loop proactive safety assurance and predictive maintenance mode with the stages of perception, analysis, decision-making, and optimization. Compared to the existing fundamental theories and platforms, which exhibit limitations and deficiencies in terms of applicability, cost-effectiveness, and implementation efficiency, the DT-IOM platform of LS-HSS highlights the following functional advantages: IoT connectivity capability, modeling & integration capabilities, virtual-real interaction capability, 2D/3D visualization capability, and service expansion capability. Similar investigations remain relatively uncommon, and this paper focuses on the fundamental methodology framework, which serves as a prerequisite for further exploration into intricate modeling, algorithm optimization, and integrated fusion. The platform proposed in this paper requires further enhancements and refinements, incorporating new technologies and methods for better application in engineering. In the future, integrating AI technology in data analysis, model optimization, and knowledge mining will further enhance the application value of the platform, leveraging its compatibility with emerging technologies and expanding functional capabilities. Consequently, this will elevate model updates, algorithm optimization, knowledge integration, and service encapsulation on the platform.

Table 1
Research and applications of digital technology for LS-HSS.

Applications focus	Research	Theories or methods
Internet of Things (IoT)	[18] (2011)	Triaxial accelerometers based on MEMS technology and vibration monitoring
Information modeling	[19] (2015)	Data mining methods based on weak association rules or non-association rules
Multi-physics modeling and simulation fusion	[20] (2018)	Numerical models and dynamic resistance analysis
IoT and Numerical simulation fusion	[21] (2018)	Strain monitoring and load distribution analysis
	[22] (2020)	Combination of numerical modeling and experimental measurements
Dynamic model correction and AI fusion	[23] (2022)	Dynamic model correction framework and surrogate model-based prediction method
	[24] (2023)	Vision-based model updating approach and failure prognostics

The main contributions of this study include: (1) establish a digital twin modeling framework of LS-HSS operation and maintenance that seamlessly integrates five levels of the physical entity, virtual entity, digital twin data, services, and connections, providing a practical digital transformation pathway, (2) explore the integration methods of data, models, knowledge, and services driven by digital twin technology to establish a closed-loop proactive safety assurance and predictive maintenance platform with the stages of perception, analysis, decision-making, and optimization, (3) the functionality and performance of the DT-IOM platform are validated in situ, effectively enhancing the digital recognition and intelligent decision-making capabilities of performance evolution characteristics for LS-HSS during long-term service.

The remaining parts of this paper (illustrated in Fig. 1) are organized as follows. Digital twin theory, implementation approach, and applications are reviewed in Section 2. A five-dimensional digital twin modeling framework of LS-HSS operation and maintenance is proposed in Section 3. Key enabling technologies and implementation methods related to the data service system, real-time mapping and online control system, dynamic health assessment system, and knowledge databases are addressed in Section 4. A comprehensive DT-IOM platform is constructed, implemented, and evaluated in Section 5, with data as the foundation, models and knowledge forming the core technologies, and the software platform acting as the service carrier. Technical advantages and performance analysis of the DT-IOM platform are provided in Section 6. Finally, conclusions and considerations for future work are drawn in Section 7.

2. Literature review

2.1. Digital twin theory and implementation approach

During a presentation on product lifecycle management (PLM) in 2003, Grieves [30] initially introduced the notion of the digital twin. With the continuous expansion of theoretical and technological advancements in physical information perception, virtual model validation, and data algorithm integration, as well as the increasing demand for application requirements [31], the development and application of digital twin technology have gradually extended from its initial focus on military and aerospace fields to a more comprehensive range of civilian sectors. Moreover, it has deeply integrated with IoT, Big Data, and AI [31]. However, one of the primary challenges in implementing digital twin technology within engineering is attaining a seamless and compatible connection between the physical and virtual domains [32]. Therefore, scholars worldwide have conducted extensive studies on digital twin theory and practical aspects to align with the emerging patterns and evolving requirements of digital twin research and application.

In the domain of modeling methods for digital twin, the primary emphasis lies in general models such as the C2PS reference model for digital twin architecture [33], object-oriented reference models [34], six-layer architecture model for digital twins [35,36], and universal

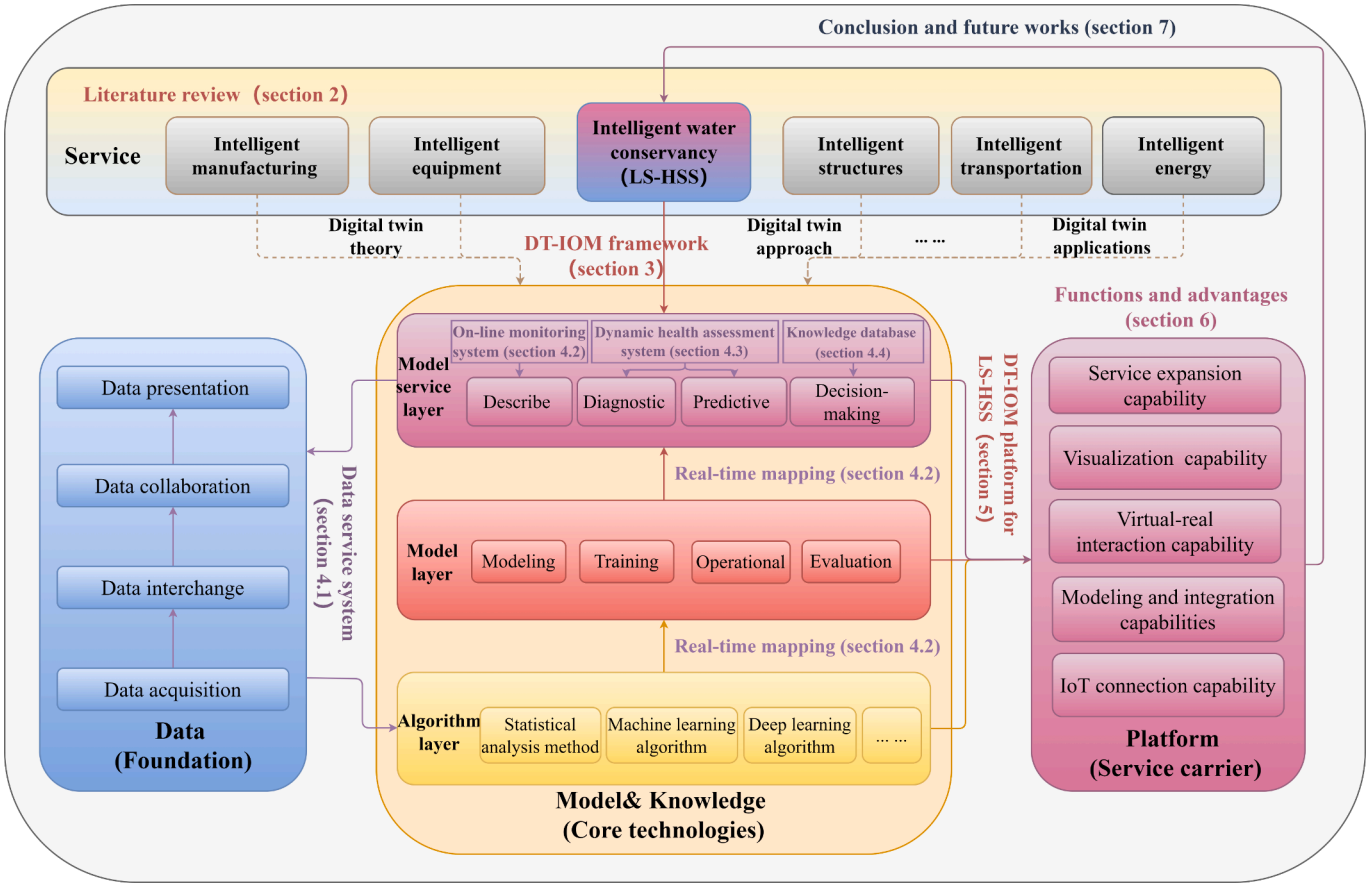


Fig. 1. The main logic and content of this study.

digital twin architecture [37]. Specialized models, including data exchange models [38,39], geometric models [40,41], and finite element models [42], are also studied. Furthermore, scholars from both domestic and international backgrounds have developed specific modeling theories and methods for various objects, such as digital workshop manufacturing processes, CNC machine tools, rotating machines, and hydraulic systems. Bao et al. [43] explored the application of digital duplicates in the manufacturing sector and proposed an approach to model these replicas specifically tailored to manufacturing settings. They also presented a theoretical framework that facilitates information integration and combines virtual reality with actual factory operations. Zhang et al. [44] introduced a five-dimensional framework for the virtual entity of a digital twin, specifically designed to optimize the utilization of robots within an intelligent manufacturing system. The primary objective of this model is to effectively address the task of reconfiguring processing resources in production while ensuring optimal efficiency. Liu et al. [45] presented a five-dimensional framework to incorporate digital twin technology into the modeling of machining processes. Luo et al. [46] developed a comprehensive modeling method for digital twin that encompasses multiple domains and explored the correlation between physical and digital space using CNC machine tools. Wang et al. [47] proposed a model for diagnosing faults in rotating machines, which utilizes a digital twin reference and incorporates a scheme for updating the model based on parameter sensitivity analysis. These advancements enhanced the practicality of the digital twin model. Wang et al. [15] provided a digital twin model with five dimensions to improve fault diagnosis in hydraulic systems. This interactive simulation model significantly enhances the accuracy of diagnosing hydraulic cylinder issues compared to non-interactive simulation models.

In the management of digital twin data, the primary areas of interest in ongoing research encompass techniques for data acquisition, methodologies for data interchange, facilitation of data collaboration, and enhancement of data presentation. A method for constructing data was introduced by Kong et al. [48], which encompasses modules for representing, organizing, and managing the data. Wang et al. [49] put forward a data acquisition method based on a Programmable Logic Controller (PLC) to enable real-time monitoring of traditional machinery. Tong et al. [50] introduced a data exchange method based on the MTConnect protocol. Schroeder et al. [39] proposed a communication strategy for exchanging and modeling twin data, utilizing AutomationML. Related research also includes blockchain-based digital twin data security sharing and management methods [27,28] and augmented reality (AR) technology-based twin data visualization methods [29]. Additionally, domestic and international researchers have explored various approaches to interactive digital twin. These encompass a framework grounded in virtual reality (VR), AR, and mixed reality (MR) for establishing connections with digital twin [51], a blockchain-based methodology for facilitating data interaction [52], an implementation of interactive digital twin utilizing model-driven software engineering [53], a natural language approach to engaging with digital twin [54], techniques for human–machine interaction in the context of digital twin [55], an integrated multi-model dynamic process modeling mechanism for enhancing interaction capabilities [56], as well as the integration between physical production systems and their corresponding digital twin [57]. Moreover, the application of digital twin-based services, including monitoring [49], assessment [58], simulation [59], forecasting [60], optimization [61], decision-making [62], and control [63], has been extensively researched by both domestic and international scholars. Furthermore, it finds applications in various fields such as aerospace engineering [64], intelligent manufacturing [61], smart cities development [65], transportation systems [58], the energy sector [62], and the equipment industry [66].

The analysis above indicates that current research in related fields of digital twin technology primarily focuses on fundamental theoretical studies, including developing frameworks and construction methods for digital twin models, data management strategies, interactive connections, and application services. However, there is a lack of emphasis on investigating DT-enabled operation and maintenance platforms to provide comprehensive integration methods of data, models, knowledge, and services and a unified reference framework. This deficiency in research impedes the further advancement of digital twins in theoretical studies and industrial applications for LS-HSS operation and maintenance.

2.2. Digital twin applications for engineering structures

In recent years, the integration and widespread adoption of digital technology in conventional sectors have led to the progressive utilization of digital twin technology for engineering structures. This has particularly resulted in significant implications for aging, antiquated, and hazardous structures. Long et al. [10] performed an extensive examination of existing literature to understand how the utilization of digital twin can enhance operational intelligence in the building construction sector. The findings indicated that existing research primarily concentrated on applying digital twin in the construction and operation/maintenance stages while neglecting its potential application in the planning/design and demolition/recovery stages. Kang et al. [6] introduced a digital twin-based approach for monitoring the health of bridges, which utilizes multimedia knowledge to synchronize virtual and real spaces. By collecting data from sensors installed in actual systems, this method can accurately reflect the current state of the bridge. As a result, it can reduce maintenance costs and prolong the lifespan of bridges. Shim et al. [7] proposed adopting a digital twin model concept to enhance decision-making reliability and optimize bridge maintenance processes in their innovative approach toward the next generation of bridge maintenance systems. Additionally, the digital twin is employed for risk monitoring and process optimization control in construction and transportation engineering domains. Kušić et al. [67] focused on simulating microscopic motorway scenarios while integrating real-time data seamlessly into system functioning. They introduced a new approach to modeling motorway traffic and demonstrated it using the Geneva motorway's digital twin model (DT-GM), which remains continuously synchronized. Jiao et al. [11] proposed a novel and effective framework for forecasting worker movements in construction sites, leveraging cloud-edge cooperation to anticipate the trajectories of workers. This approach ensures data privacy and minimizes computational load while achieving accurate results.

Meanwhile, the issues of safety monitoring and health management for engineering facilities and equipment are becoming increasingly prominent as the service life of hydraulic engineering projects extends [8]. Therefore, digital twin technology is progressively integrated into operational and maintenance procedures, guaranteeing all-encompassing safety across the lifespan. Zhang et al. [8] proposed a novel online learning algorithm called AF-OS-ELM-MVE to address dam deformation monitoring challenges. Shi et al. [12] put forward the utilization of the digital twin approach for managing hydraulic engineering operations to attain intelligent operational procedures, accurate regulation, and reliable maintenance in hydraulic engineering. Li et al. [68] introduced a digital twin-based intelligent monitoring system architecture to expand the scope of digital twin applications and enhance the overall lifespan safety management of dam operations. Wang et al. [14] proposed a maintenance model based on digital twin for electromechanical devices to predict health status. Liu et al. [16] described an innovative framework for digital twins, which can be a robust tool for assessing the present conditions of canals and forecasting future trends. This framework significantly improves security and efficiency in extensive water diversion projects. With the advancement of information technology, there has been continuous improvement in the digitization, networking, and intelligence levels within hydraulic engineering. The utilization and exploration of digital twin technology are experiencing significant growth. Earlier studies have explored the application and effects of the digital twin in hydraulic engineering, primarily focusing on areas such as dam safety monitoring, electromechanical devices, and water resource management. Insufficient attention has been given to the digital transformation pathway of conventional operation and maintenance for LS-HSS and the integration methods of data, models, knowledge, and services driven by digital twin technology. In addition, the actual operating conditions and demands for different engineering structures in situ are complex, discrepant, and changing. Therefore, a comprehensive DT-driven LS-HSS intelligent operation and maintenance (DT-IOM) platform is essential.

In summary, the current methodologies do not encompass a comprehensive framework of digital twin modeling for LS-HSS operation and maintenance. Moreover, exploring integrating methods of data, models, knowledge, and services driven by digital twin technology is essential. Subsequently, establishing a comprehensive DT-driven intelligent operation and maintenance (DT-IOM) platform becomes crucial in promoting the digital transformation of conventional LS-HSS operations. We believe that this study will provide valuable references in the future.

3. Framework of digital twin-driven LS-HSS operation and maintenance

The digital twin represents a significant technological advancement enabling seamless interaction and integration between physical and virtual environments, thereby driving widespread adoption across diverse domains. In Grieves's research [30], the initial concept of digital twin encompassed physical entities, virtual entities, and their interconnectedness. However, the emergence of digital twin models has led to new trends and demands because of application needs and advancements in theoretical technologies. In this study, the intelligent operation and maintenance framework for LS-HSS is designed with a five-dimensional digital twin model [32], including physical entity, virtual entity, DT data, services, and connections (see Fig. 2). The digital twin modeling framework for LS-HSS operation and maintenance focuses on integrating data, models, knowledge, and services.

Then on this basis, to provide a pathway for digitally transforming conventional LS-HSS operation and maintenance, a DT-driven LS-HSS intelligent operation and maintenance (DT-IOM) platform features real-time mapping, closed-loop control, dynamic health assessment, and intelligent decision-making is constructed and applied. The platform architecture comprises four layers: physical entity layer, digital twin layer, service layer, and connection layer, as shown in Fig. 3. The perception and processing of data offer crucial information for real-time bidirectional mapping and interoperability. The connection in real-time between the digital twin layer and service layer offers data assistance for immediate control directives. Ultimately, the intelligent perception and optimal control of the LS-HSS can establish a closed-loop proactive safety assurance and predictive maintenance mode with the stages of perception, analysis, decision-making, and optimization.

3.1. Physical entity layer

The physical entity layer comprises essential elements of LS-HSS and subsidiary facilities: major structures (panel, gate leaf, girder, radial arm, etc.), hoists (hydraulic station, hydraulic cylinder, control system, etc.), and hydraulic structures (lock chamber, sluice pier, wing wall, etc.). These components enable the LS-HSS to undertake vital functions such as flood management, water supply for diverse purposes, irrigation facilitation, power generation, and navigation in hydraulic engineering.

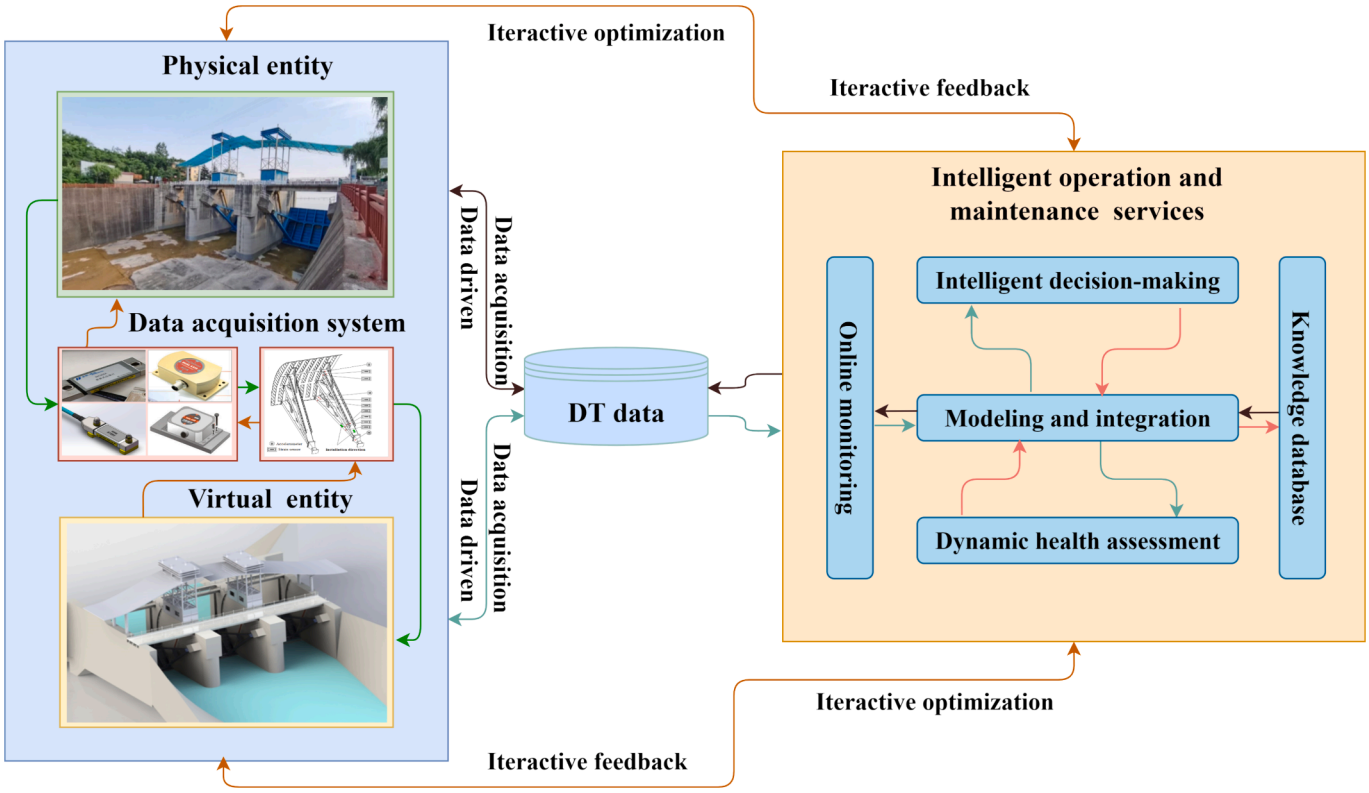


Fig. 2. DT-driven LS-HSS intelligent operation and maintenance under the five-dimensional framework.

3.2. Digital twin layer

The data platform, model platform, and knowledge platform constitute the digital twin layer. The data platform has functions, including data aggregation, management, mining, and services. It incorporates functional components such as PLCs, industrial computers, and sensors for acquiring data and facilitating communication. These components enable the platform to achieve data acquisition, interaction, and real-time monitoring between the tangible and digital domains. It integrates the operation and maintenance data and the digital twin data to establish a cohesive foundation of unified information. The operation and maintenance data mainly include design, monitoring, and maintenance data. The data of the digital twin encompass information on mechanism analysis, mathematical statistics, simulation analysis, iterative optimization, and decision-making. Once cleaned, these diverse data sets are promptly stored in the database to supply the application layer with real-time data support.

Simultaneously, the model platform is a digital reconstruction of LS-HSS, encompassing a virtual depiction of the major structures, hoists, hydraulic structures, etc. Based on a three-dimensional geometric model and employing intelligent sensor recognition, mechanical analysis, and simulation fusion, a digital twin model for LS-HSS is ultimately constructed. Hence, the digital twin model possesses identical geometric measurements, positional data, and behavioral information as tangible equipment to ensure efficacy and accuracy. Its objective is to establish a more realistic connection between operational and maintenance tasks in both physical and virtual domains. Meanwhile, a knowledge platform, including standard specifications, expert knowledge, contingency plans, maintenance experiences, rules, etc., is established to support intelligent decision-making. The data platform, model platform, and knowledge platform collectively constitute the geometries, physical properties, behaviors, and rules of the digital twin model. Additionally, the digital twin layer facilitates the synchronization of data and model platforms through integration and interconnection. This enables seamless integration, promotes analysis, provides virtual mapping in real-time, provides statistical insights, and enhances efficiency in digital operations and maintenance activities.

3.3. Connection layer

The connection layer (CN) enables seamless communication between different platform components, including the connection from the physical entity layer to the digital twin layer and the service layer. In this procedure, real-time collection of physical entity data is facilitated by utilizing various sensors, embedded systems, data acquisition cards, etc., with data as a medium. The acquired information is subsequently transmitted to the data platform via protocols like MTConnect, OPC-UA, and MQTT [69]. Following processing on the data platform, the processed instructions or information can be relayed to the physical entity for operational optimization through protocols such as OPC-UA, MQTT, and CoAP [69]. At the same time, real-time storage of simulation-related and service data is facilitated in the data platform through database interfaces like JDBC and ODBC [69]. Simultaneously, dynamic utilization of fusion data, associated data, historical data, rule data, and commonly used algorithms and models enables real-time simulation. Furthermore, it provides support for service operation and optimization.

3.4. Service layer

The current focus in the field of hydraulic engineering in China is to accelerate the development of a digital twin system for water resources management, emphasizing advancing its construction to possess capabilities such as forecasting, issuing early warnings, conducting simulations, and drafting contingency plans [12]. For the subsequent seamless integration of intelligent operation and maintenance services of LS-HSS into the digital twin-enabled watershed construction, this paper introduces a digital twin-based intelligent operation and maintenance

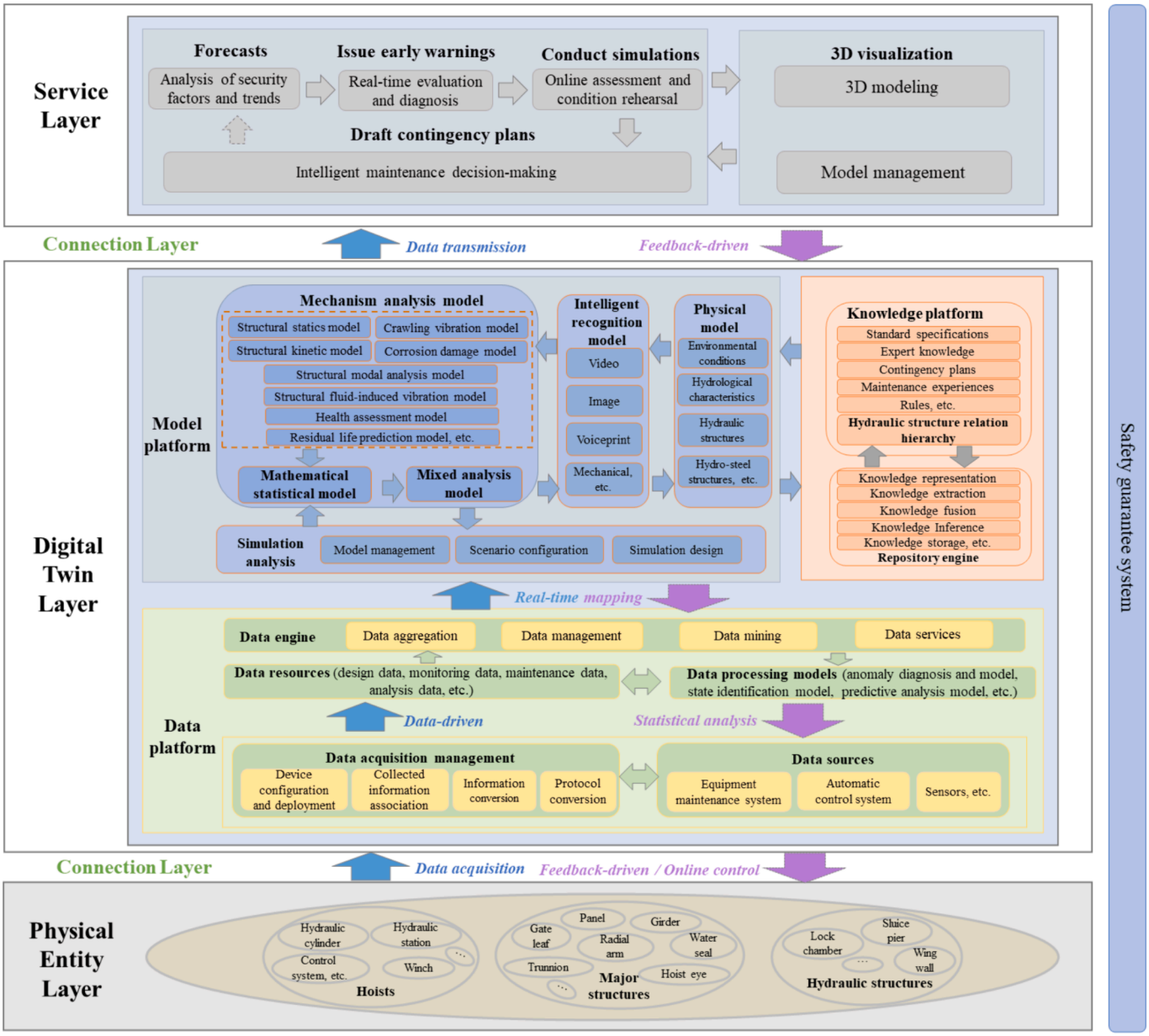


Fig. 3. Architecture of DT-driven LS-HSS intelligent operation and maintenance platform.

service workflow, as shown in Fig. 4. The objective is to integrate these conventional approaches for intelligent operation and maintenance, encompassing disparate entities, fragmented information, and limited system integration, into a cohesive closed-loop system while enhancing precision and efficiency.

Forecasting various safety elements of LS-HSS (e.g., water level, velocity, flow, temperature and humidity, wind speed, stress, vibration, and inclination) is essential for predicting and anticipating potential risks in advance. Early warning acts as a forefront, offering guidance to frontline water conservancy personnel by promptly notifying them about various risks and potential secondary disasters (e.g., structural damage, excessive stress, abnormal vibration, irregular posture, corrosion, and leakage). It guides valuable insights for conducting simulations. Simulation before the actual event is of utmost importance in the digital twin layer, as it entails accurately replicating equipment operations and maintenance within specific event scenarios. This ensures the precision of the constructed model and enables simulation and emulation of equipment utilization and maintenance in design, planning, or future forecasting scenarios. By promptly identifying issues, scientific formulation and solutions can be optimized. The ultimate objective is to formulate contingency plans encompassing the duration of equipment usage, sequential arrangements, regulations, diverse maintenance measures, and implementation processes. The interconnectedness and progressive nature of the service functions serve as both the beginning and end point for the digital twin of LS-HSS.

In summary, the service layer plays a crucial role in the DT-IOM platform by encompassing the operational and maintenance information system and the 2D/3D visualization system. The primary purpose of the information systems for operation and maintenance is to manage equipment effectively. This involves analyzing security factors and trends, making forecasts, providing real-time evaluation and diagnosis to issue early warnings, conducting online assessments and simulations of operating conditions, and facilitating intelligent decision-making for maintenance by drafting contingency plans. Using 2D/3D visualization in operation and maintenance information systems facilitates the real-time synchronization of operations and enables multi-angle monitoring of physical activities through DT data in a virtual environment. As a result, the analysis of vast data from the digital twin layer enhances

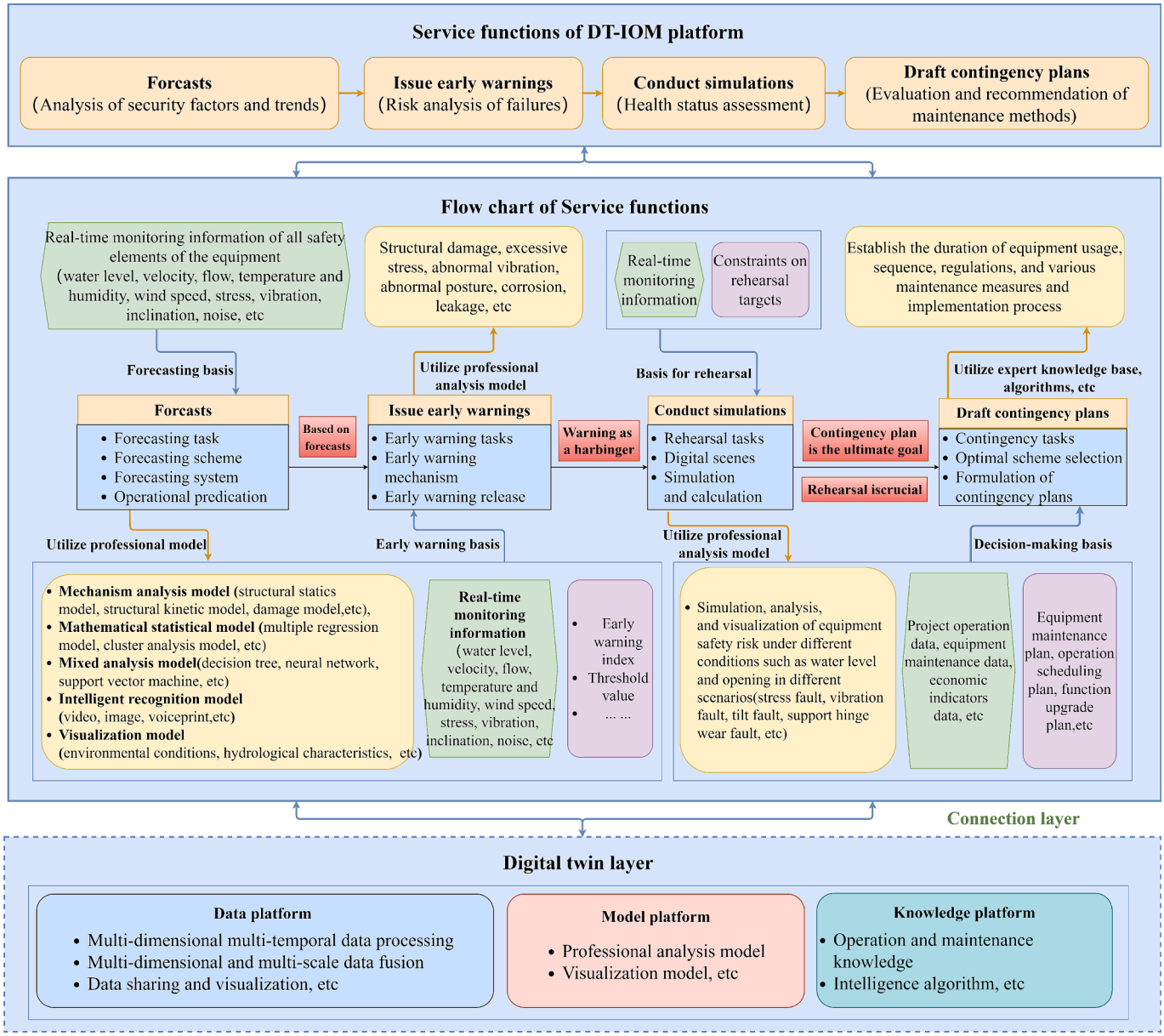


Fig. 4. Workflow of service layer of the DT-IOM platform.

decision-making for control processes, enabling optimal outcomes. The primary components of the 3D visualization system are centered around 3D modeling and managing models. Ultimately, these efforts lead to realizing crucial functionalities within the DT-IOM platform to establish a closed-loop proactive safety assurance and predictive maintenance system with the stages of perception, analysis, decision-making, and optimization.

4. Key enabling technologies

Successful implementation of a DT-IOM platform for LS-HSS necessitates utilizing DT-driven data, models, knowledge, and services integration methods. The primary focus is overcoming technical obstacles related to data acquisition and analysis, real-time mapping, online monitoring, health assessment, etc.

4.1. Data service system of LS-HSS intelligent operation and maintenance platform

Data plays a crucial role in facilitating and serving as a vital element within the DT-IOM platform for LS-HSS. The virtual models can comprehend the condition, forecast, approximate, and scrutinize dynamic alterations of physical entities by utilizing sensory data obtained from the physical entities [66]. Before receiving data services, data processing involves a series of stages, such as data procurement, filtration, examination, extraction, transfer, retention, presentation, and utilization [69]. The structure of the data service system for the DT-IOM platform is depicted in Fig. 5, consisting of three layers: perception, processing, and application.

The data for this research is collected from two primary sources: routine maintenance data of the LS-HSS management system and operational status data of the online monitoring system. It can be rapidly gathered via digital inspection equipment (handheld computers or PDA devices), sensors, control systems, and other dynamic data acquisition instruments. The acquisition of crucial data from physical entities in the

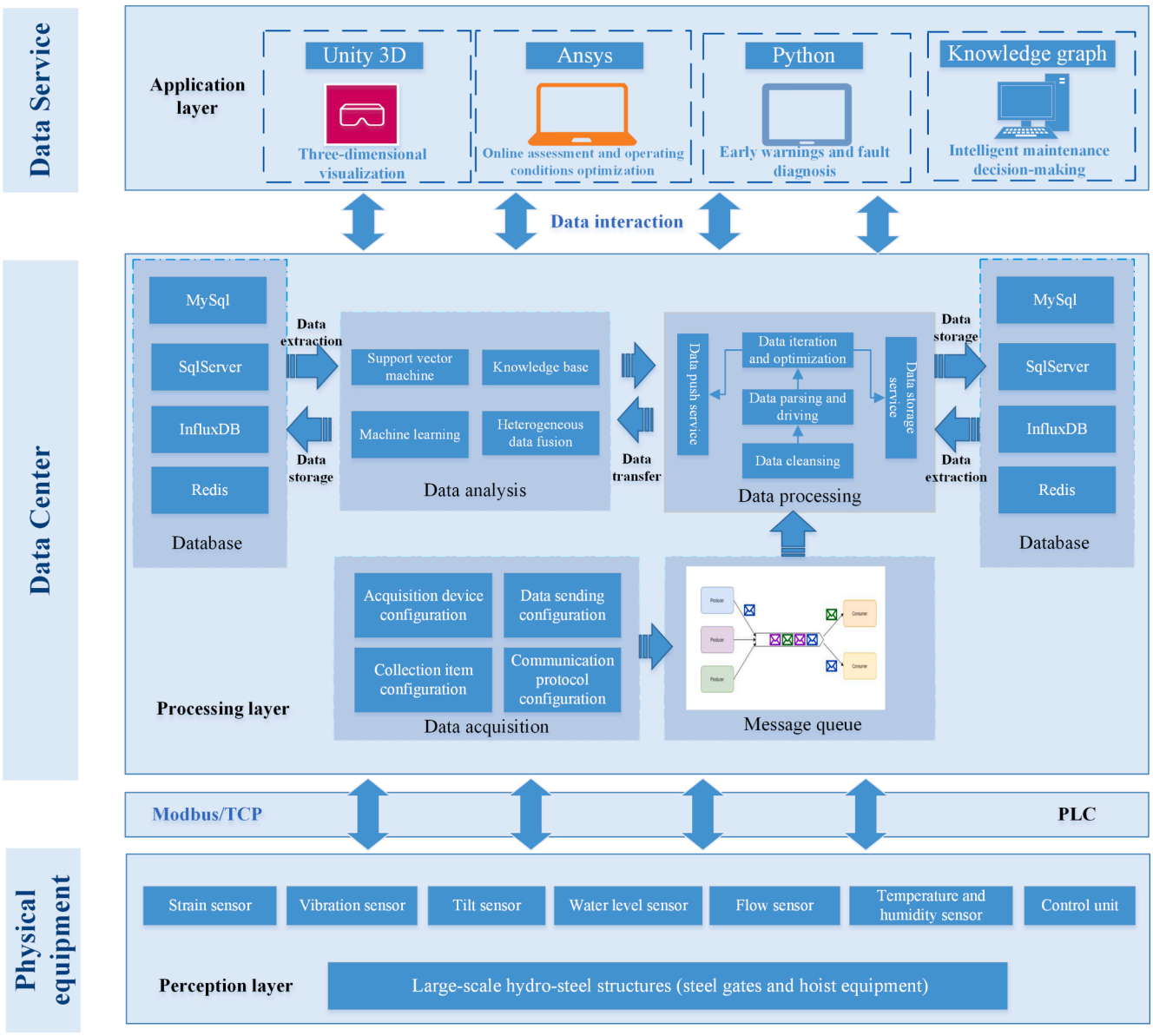


Fig. 5. Architecture of data service system for LS-HSS intelligent operation and maintenance platform.

DT-IOM platform is presented in Table 2. One of the specialized instruments available for capturing and assessing the operational condition of LS-HSS is the dynamic signal testing and analysis device manufactured by Donghua Testing Technology Co., Ltd. in China. This instrument facilitates measuring and evaluating vibration, strain, and attitude parameters.

Fig. 6 shows the data acquisition system for the DT-IOM platform. This system enables the online monitoring of various aspects, including static stress in the primary components, dynamic stress, dynamic responses, operational posture, and the operational status of control units and environmental conditions. The acquisition frequency can be fine-tuned up to 200 times per second, effectively fulfilling the criteria for real-time correlation between physical entities and digital representations. Additionally, the routine maintenance data are collected through regular inspections and manual entry of historical maintenance data, creating corresponding datasets containing condition-specific data (such as environmental parameters and recent maintenance data.). These DT-IOM resource data are sent to the intelligent gateway and uploaded standardized using a fixed Modbus/TCP communication protocol.

The multi-source diverse data undergoes a series of procedures, including data purification, extraction, organization, refinement, and enhancement, aiming to eliminate duplicated, deceptive, inaccurate, and inconsistent information. Different techniques, such as heterogeneous data integration, support vector machines, machine learning algorithms, and other approaches, are employed to conduct a more in-depth data analysis. This aids in the extraction of crucial information and facilitates the generation of novel insights. These extensive amounts of information, encompassing simulation, operational, environmental, and other process data, are consolidated and stored within the database.

The data service capability can be regarded as a metric for assessing the adaptability and intelligence exhibited by DT-IOM. The range of specialized solutions provided by the data services encompasses various functionalities such as visualization, optimization, and diagnosis. Visualization service is based on high-fidelity 3D modeling technology, which constructs three-dimensional equipment exteriors and interior models. It integrates engineering structures and sensor monitoring device information to achieve access, interaction, and integrated display of monitoring data, analysis data, and warning information. It also supports human–computer interaction through mobile internet and intelligent terminals in three-dimensional virtual scenes. Based on water

Table 2
Key data acquisition of LS-HSS intelligent operation and maintenance platform.

Data acquisition type	Data acquisition equipment	Data type	Data source	Communication protocol
Vibration of gate	Vibration sensor	Float	Dynamic data acquisition instrument	Modbus/TCP
Stress and strain of gate	Strain sensor	Float	Dynamic data acquisition instrument	Modbus/TCP
Tilt Angle of gate	Tilt sensor	UINT	Dynamic data acquisition instrument	Modbus/TCP
Gate opening meter	Control unit	Float	PLC	Modbus/TCP
Opening and closing force of the hoist		Float	PLC	Modbus/TCP
Opening and closing speed of the hoist		Float	PLC	Modbus/TCP
Current and voltage of hoist		Float	PLC	Modbus/TCP
Water level	Water level sensor	UINT	Sensor	Modbus/TCP
Flow velocity	Flow sensor	UINT	Sensor	Modbus/TCP
Ambient temperature and relative humidity	Temperature and humidity sensor	UINT	Sensor	Modbus/TCP

resource scheduling objectives, scheduling rules, and upstream/downstream flow conditions, the optimization service dynamically simulates (e.g., ANSYS). It calculates the operational status of equipment within a foreseeable period. Integrating actual operating conditions with simulation results from running virtual scenarios optimizes predictive maintenance plans for the equipment. Diagnosis service provides functions such as online/inspection status monitoring, early warning, and fault diagnosis analysis using artificial intelligence algorithms. In addition, knowledge graph technology facilitates the structured management of operation and maintenance knowledge, encompassing rule libraries, model libraries, case libraries, etc. This enables convenient and efficient maintenance, reuse, and querying of operation and maintenance expertise and information. Meanwhile, real-time updates and interactions between the database and data service can facilitate comprehensive monitoring and traceability management throughout the operation and maintenance process.

4.2. Real-time mapping and online monitoring system of LS-HSS intelligent operation and maintenance platform

Fig. 7 illustrates the virtual-real interaction mechanism. The physical entity performs the data acquisition process in real-time and then transfers the acquired data to the data service system for processing and storage. The integration of virtual and physical is achieved through continuous access to the data service system, enabling the digital twin platform to acquire real-time data. The digital twin platform encompasses services including real-time data-driven operational status monitoring, online assessment, rehearsal of LS-HSS operating conditions, and intelligent maintenance decision-making.

The services are achieved through a combination of methods, including analysis of determinants impacting health status, multi-sensor information fusion data acquisition, real-time safety assessment, and knowledge-driven process control optimization. This serves the dynamic evaluation and accurate prediction of safety elements during LS-HSS operation. The framework for real-time data-driven operating status monitoring is as follows in Fig. 8 and is comprised of the subsequent procedures:

(1) Analysis of determinants impacting health status.

The design and processing quality influence the strength, stiffness, and stability of LS-HSS. Meanwhile, these mechanical properties can be impacted by environmental conditions, hydraulic conditions, equipment operation, and maintenance, leading to performance degradation and potential safety hazards. The conventional LS-HSS operation and maintenance often takes insufficient consideration of the relationship between various indicators, resulting in a single-dimensional evaluation (such as structural stress–strain or vibration) of equipment operation safety. As a result, the accuracy and efficiency of identifying safety risks and health conditions during equipment operation are affected.

Therefore, in this paper, the health evaluation indexes of LS-HSS are established through a comprehensive analysis of safety risk factors in the design, manufacturing, operation, and maintenance processes. These primarily include Mechanical Performance Indicators (MPI), such as strength, stiffness, and stability; Working Performance Indicators (WPI), which encompass vibration, tilt angle, control parameters, hydrodynamics, and environmental parameters; Operational Maintenance Indicators (OMI), which cover regulations compliance, operational proficiency, and the maintenance and repair status of LS-HSS.

The interrelated and non-orthogonal characteristics of the indicators above in both structural mechanics and data science dimensions necessitate a unified analysis to facilitate dynamic health assessment and enable intelligent decision-making. To fulfill this requirement, we define the corresponding data tables (Dt) for MPI, WPI, and OMI as follows:

$$Dt = \{Tablename, Fieldname, Fieldtype, Fieldlength, Mandatory, Defaultvalue, Remarks\} \quad (1)$$

Besides, the data identification (Di) for the specific indicators is defined as follows:

$$Di = \{Acquisitiontime, Itemid, Datavalue, EquipmentID, AcquisitionobjectID, sensorID, Unit\} \quad (2)$$

(2) Multi-sensor data acquisition

As mentioned in section 4.1, the data for this research is collected from two primary sources: routine maintenance data of the LS-HSS management system and operational status data of the online monitoring system.

(3) Real-time evaluation and analysis.

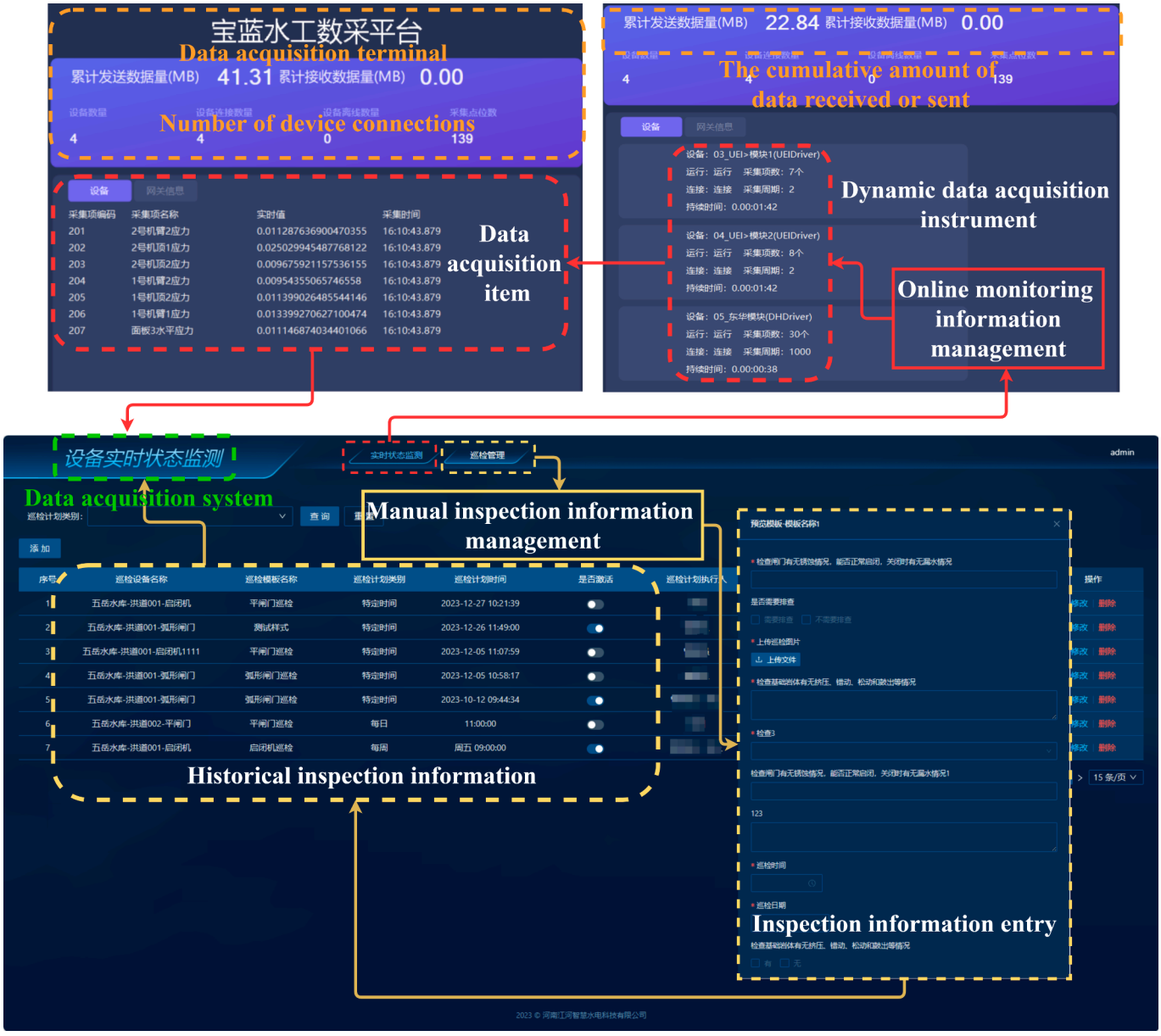


Fig. 6. Data acquisition system for LS-HSS intelligent operation and maintenance platform.

For *MPI*, dynamic comparative analysis models are constructed by combining online monitoring data with data from the digital twin simulation for dynamic assessment, as follows:

$$OperatingCondition = \begin{cases} Safe : \sigma \le \sigma_f \\ Relativelysafe : \sigma_f \le \sigma < [\sigma] \\ Dangerous : [\sigma] \le \sigma < \sigma_s \\ Destructive : \sigma \ge \sigma_s \end{cases} \tag{3}$$

Where σ refers to the actual observed value, σ_f refers to the calculated value from the full-load digital twin model, $[\sigma]$ refers to the allowable strength, and σ_s refers to the material yield strength.

For *WPI* and *OMI*, machine learning methods (support vector machine model, unsupervised learning model, etc.) are developed. By incorporating expert knowledge and standard specifications, it is built on historical operating condition characteristics and operational datasets to enable dynamic assessments of LS-HSS operating conditions.

Additionally, the data collected by general sensors usually contains many signal anomalies. These signal anomalies are unrelated to structural health deterioration but originated from equipment abnormalities [70]. To solve this problem, the identification of false alarms is necessary. Firstly, to mitigate false alarms caused by sensor damage or abnormal environmental conditions, a predefined range for each parameter is established through statistical methods to identify false alarms [71]. If the alarm duration is less than 30 s without repeated occurrences based on the experience, no alarm will be triggered to eliminate false alarms. Otherwise, the same parameters from the same structural components (for example, σ and σ_{DT} calculated by the digital twin model under the same operating conditions) or additional parameters from the same structural components (for example, MPI, WPI, and OMI) are utilized for state verification in case of an alarm event. If the deviation exceeds 30 % or more than 80 % of the additional parameters indicate normal conditions, no alarm will be triggered to eliminate false alarms. Simultaneously, manual investigation of anomalies is conducted to address any issues related to sensor false alarms promptly. The future entails conducting further optimization research on the thresholds or methods for determining relevant anomalies through data accumulation, continuous integration of analysis, and on-site testing. This will significantly enhance the accuracy of false alarm identification and

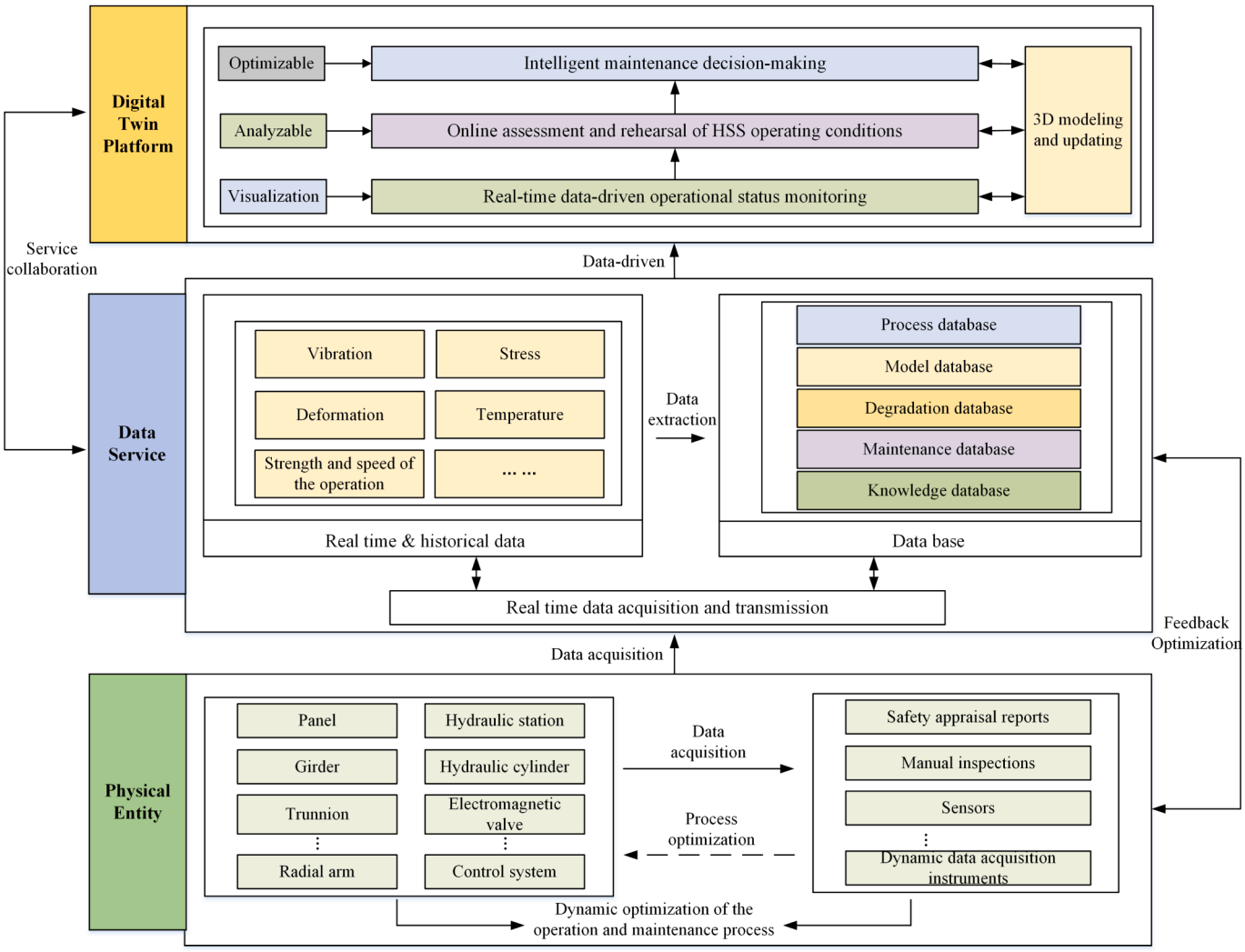


Fig. 7. Architecture of real-time mapping and online monitoring system for LS-HSS intelligent operation and maintenance platform.

judgment in DT-IOM platform application processes.

(4) Knowledge-driven process optimization

Upon real-time assessment and analysis of LS-HSS operating conditions, the system identifies any decline in LS-HSS performance or potential safety risks. In response, it optimizes operational control strategies based on the knowledge repository of operational control methods and maintenance methods. If necessary, proactive maintenance is executed to ensure the real-time performance of the LS-HSS. The operational control and maintenance methods and their effectiveness are documented and stored in the knowledge repository. This repository is dynamically updated to maintain stable and efficient operation. The DT-IOM stands out in its efficient maintenance process data sets, dynamic multidisciplinary simulation, and real-time interaction in virtual environments compared to conventional maintenance procedures. It establishes a virtual-real interaction and coordination mechanism to achieve dynamic data interaction and efficient collaborative objectives in the DT-IOM of LS-HSS.

4.3. Dynamic health assessment system of LS-HSS intelligent operation and maintenance platform

Considering the effects of changing circumstances over time and maintenance procedures on the structure and operational efficiency of the LS-HSS, it is essential to conduct health assessments throughout the lifecycle to ensure its reliability and safe operation. Fig. 9 illustrates the process of dynamic health assessment of the DT-IOM platform. As in the case of a radial gate, the process begins in the virtual space by constructing a finite element model that includes geometric and physical data based on the original design drawings. During this construction, various factors, such as external driving forces, environmental disturbances, operational mechanisms, constraint relationships, and behavior and rules, influence the LS-HSS during its service life. Therefore, response surface methodology [72] is incorporated into the finite element model to facilitate model updating, ensuring its accuracy following the operational process. The flow chart depicting the response surface methodology for the dynamic health assessment system can be observed in Fig. 10.

(1) The main parameters that influence the response of LS-HSS, including material parameters (elastic modulus, density, Poisson's ratio) and geometric parameters (length, thickness, etc.), should be initially identified. The stress and vibration data obtained from real-time monitoring technology in Sections 4.1 and 4.2 should be the target function for modifying the finite element model. The optimization objective is to minimize the disparity between the calculated values of the modified finite element model response and experimental measurements.

(2) The Central Composite Design (CCD) method is utilized for the experimental design of parameter samples, generating sample points. The F-test is employed to analyze the parameters' significance, and higher-significance parameters are selected for response surface fitting.

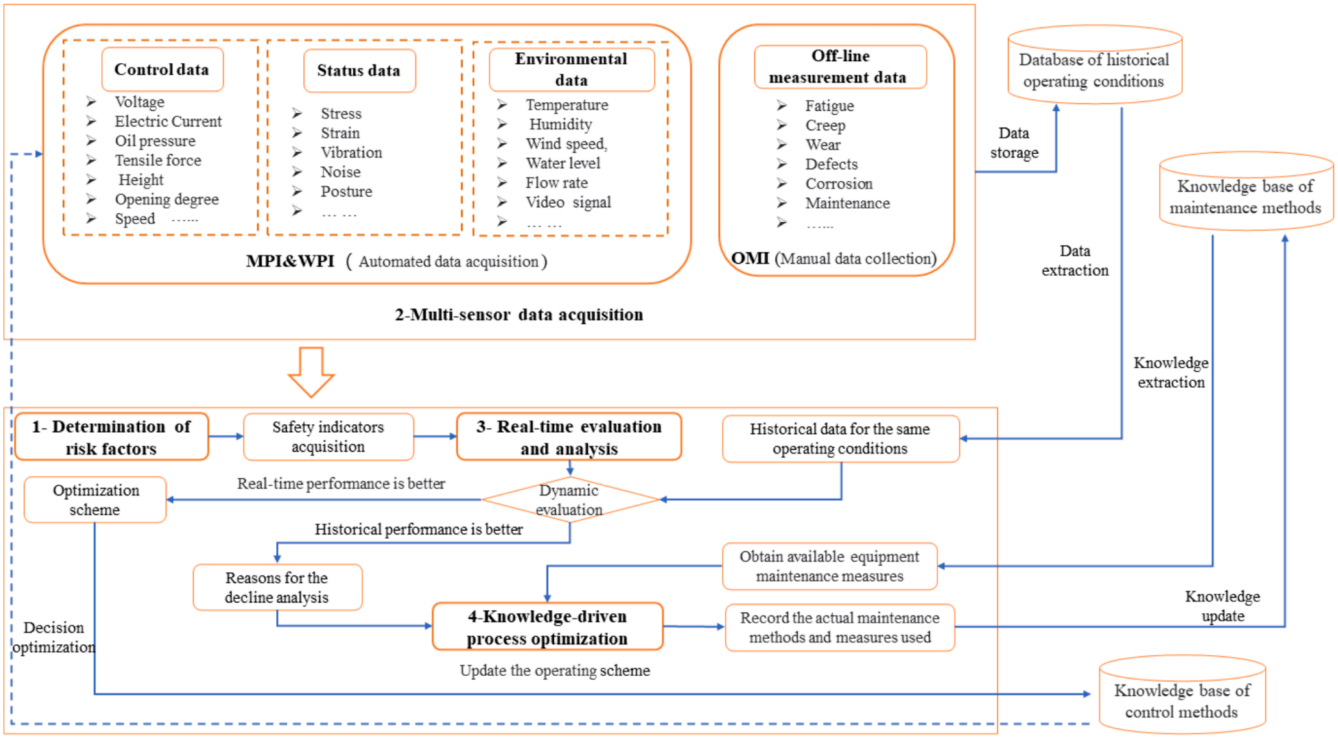


Fig. 8. The flowchart of real-time data-driven operating status monitoring.

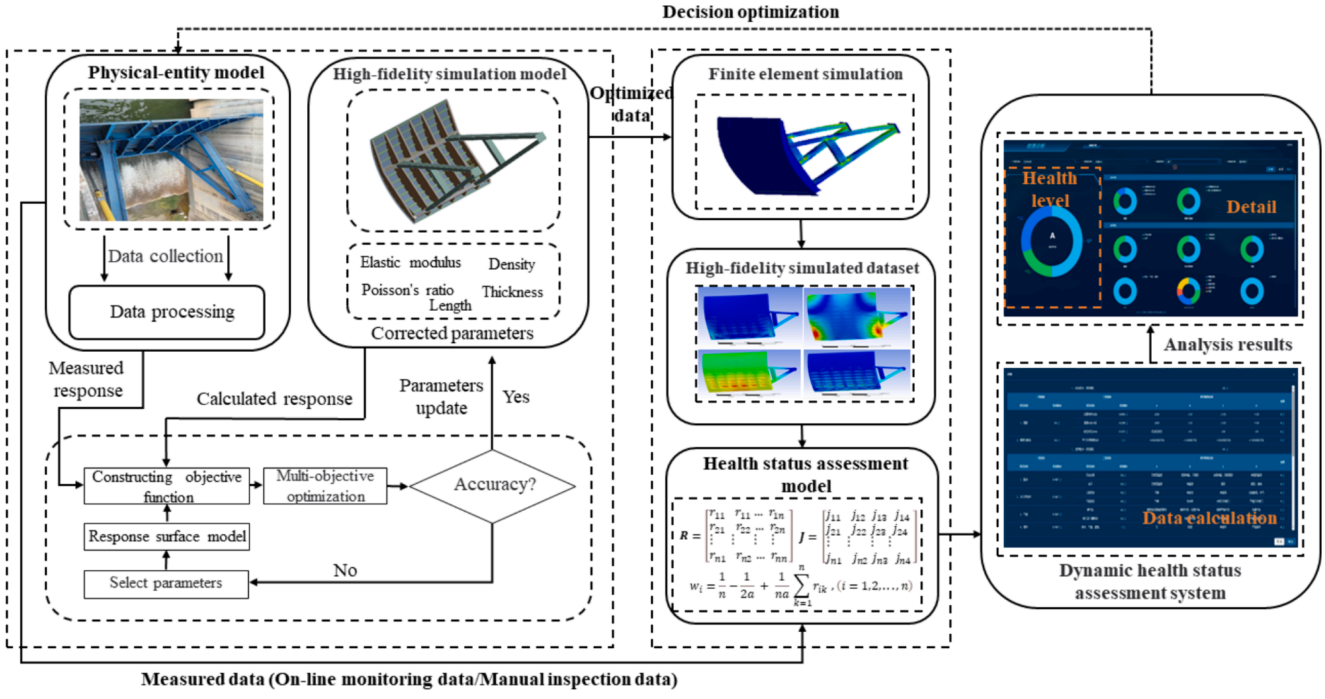


Fig. 9. The process of dynamic health assessment system.

The response surface models encompass a range of types, including polynomial, nonlinear function, power function, and BP neural network models. The quadratic polynomial response surface model is employed in this paper as follows [73]:

$$y = \beta_0 + \sum_{i=1}^{k} \beta_i x_i + \sum_{i=1}^{k} \sum_{j=1}^{k} \beta_{ij} x_i x_j + \sum_{i=1}^{k} \beta_{ii} x_i^2 \tag{4}$$

Where $x_i \in [x_i^l, x_i^u]$, $i \in (1, k)$, x_i^l, x_i^u are the range of values for design parameters x_i, and $\beta_0, \beta_i, \beta_{ij}, \beta_{ii}$ are regression coefficients.

Next, the response surface model's accuracy is evaluated using the following method for assessing its precision:

$$R^2 = 1 - \frac{\sum_{i=1}^{N} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{N} (y_i - \overline{y}_i)^2} \tag{5}$$

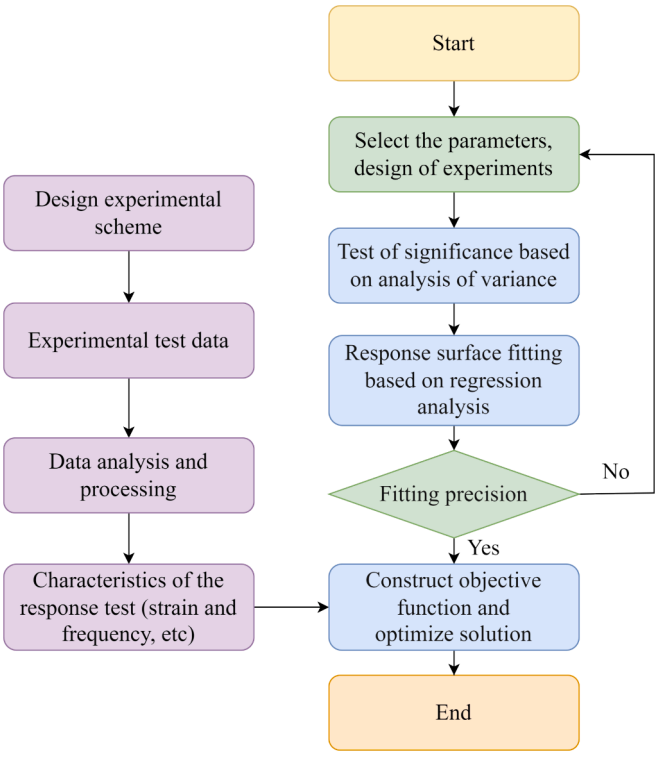


Fig. 10. The flowchart of the response surface methodology.

Where y_i represents the finite element calculation results, $\hat{y}_i$ denotes the response surface calculation results, N is the number of test points in the design space, and $\overline{y_i}$ represents the average value of finite element calculation results. The coefficient of determination R^2 ranges from 0 to 1. A higher R^2 value indicates that the obtained regression response surface model is closer to reality; a lower R^2 value, especially when it reaches a certain extent, suggests that a new experimental design should be conducted.

(3) Finally, the modified objective of the finite element model is defined as follows:

$$min\sqrt{\sum_{i=1}^{m}\beta_i[f_{Ei}(X)-f_{Ai}(X)]^2}, VLB \le X \le VUB \tag{6}$$

where X represents design parameters, f_E is the experimental value, and f_A is the analytical value. β_i denotes weights. Due to the randomness of sample points, their contribution varies when constructing the objective function. To improve the accuracy of adjustments, sample points deviating from the objective function are assigned smaller weights, while those closer to it carry greater importance in terms of weightage. *VLB* and *VUB* represent the range of parameter design space.

This approach fits response surfaces to sample points within the variable design space to accurately represent the actual behavior. The response surface model replaces the finite element model for rapid iterations. Building upon the constructed response surface model, the physical entity of the radial gate is optimized using real-time sensor data as the objective function. This results in a high-fidelity finite element model that aligns with the operational behavior and can be used as a reference model for future analysis of damages and evaluation purposes. Next, finite element analyses are performed using the high-fidelity finite element model of the radial gate as a foundation. The radial gate's behavior and rule model data are inputs, and their geometric and physical model data are generated as outputs. This process yields a more precise operation digital twin dataset to facilitate LS-HSS health assessment and optimize operational decision-making and management. In addition, the present study utilizes the radial basis function (RBF) neural network to construct a surrogate model based on the stress simulation dataset. This enables a preliminary dynamic simulation test of the equipment operation process. The procedural steps involved are as follows:

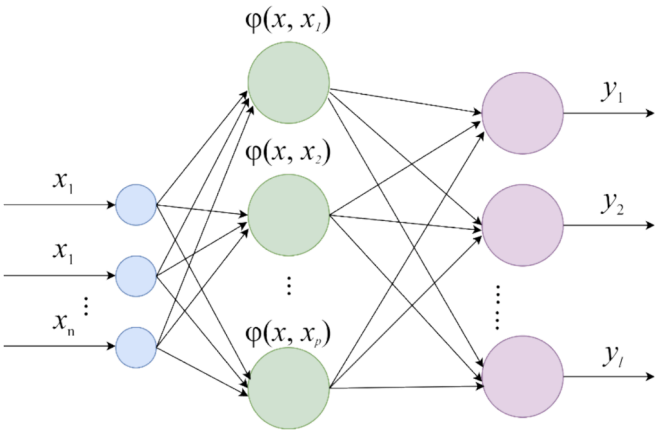


Fig. 11. Structure of typical RBF neural network.

(1) The offline simulation analysis module is utilized to compute and export the analytical data of the gate under various opening or closed states at specific water levels (taking stress data as an example). The stress values of each node corresponding to the gate under different operational conditions are represented by a four-element array (x, y, z, Mpa), with each CSV file representing the stress data of all nodes for a specific operating condition position (e.g., 1.csv represents the stress data of all nodes for the gate at opening position 1).
(2) This model is essentially a neural network with a single input (opening position or current water level) and multiple outputs (stress values of all nodes), which incorporates the actual operating conditions of LS-HSS. The structure of a typical RBF neural network is illustrated below (see Fig. 11).
(3) The input to each neural network is either the opening or current water level (X), and it generates stress values for all nodes under that specific operating condition. The training process in a Python environment is illustrated in Fig. 12, where each state corresponds to a multidimensional matrix that stores stress information.
(4) The TCP server and client are developed separately in Python and Unity environments. The program code in the Python server invokes the model and dynamically acquires input and output values, thereby achieving seamless integration of data, models, and services.

Then, considering the interaction between health indicators, a health assessment model based on the fuzzy analytic hierarchy process (FAHP) is employed to accomplish the dynamic health evaluation. The model's construction process is outlined as follows:

(1) Develop the LS-HSS health assessment indicator system and establish a hierarchical structure model for comprehensive diagnosis of LS-HSS health [74], as illustrated in Fig. 13. Organize the decision problem into different levels based on the hierarchy, starting from the objective level, and progressing to the criterion level, sub-criterion level, and finally reaching the index level.

(2) Method for acquiring and quantifying indicator parameters: The above indicators are derived from expert scoring, manual inspection input, online monitoring, or simulation analysis methods as outlined in Sections 3.1 and 3.2. The grade standards for each indicator are determined primarily based on its characteristic traits and the extent of its impact on LS-HSS health. These indicators are divided into four grade

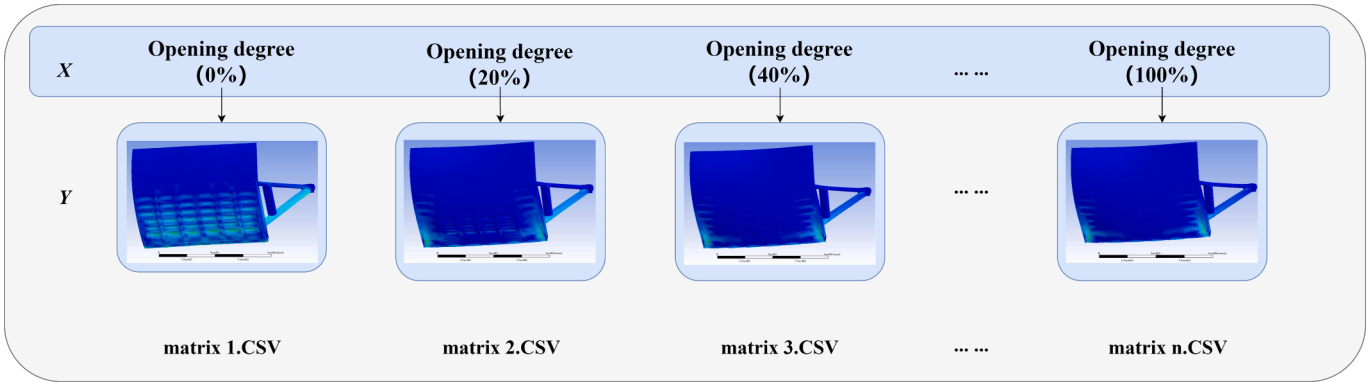


Fig. 12. Training process of the predictive model.

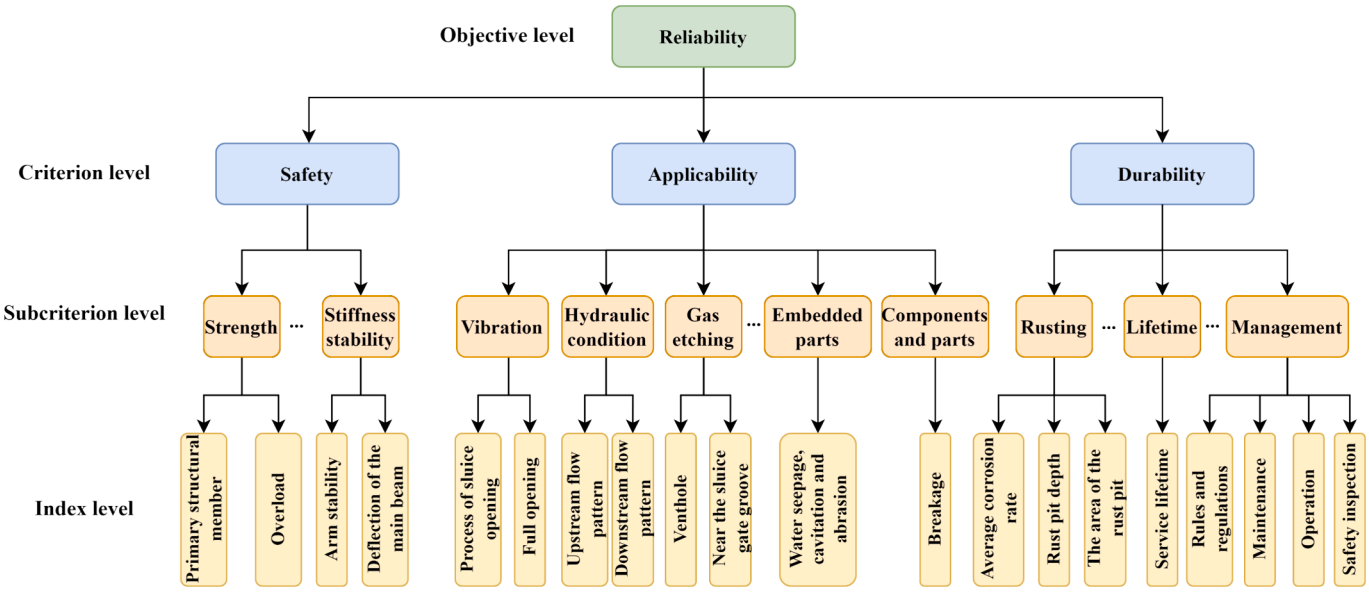


Fig. 13. The hierarchical structure model for comprehensive health assessment of LS-HSS.

standards (A, B, C, and D) to facilitate corresponding discrimination [75].

(3) The criteria and sub-criteria weighting process [76] involves experts comparing each criterion and sub-criterion pairwise using fuzzy linguistic descriptions, constructing a fuzzy judgment matrix. By utilizing the commonly used 0.1 to 0.9 scale method to establish numerical indicators, along with the formula provided below for defining the fuzzy complementary matrix, the scoring results of pairwise comparisons that meet the fundamental requirement of a fuzzy complementary matrix are recorded as $\boldsymbol{R}$ in the final scoring matrix, which is represented as follows:

$$0 \le S_{ij} \le 1, S_{ii} = 0.5, S_{ij} + S_{ji} = 1, (i, j = 1, 2, \cdots, n) \tag{7}$$

Where i and j represent influential factors, S_{ij} denotes the relative importance ratio of factor i to factor j, while S_{ji} represents the relative importance ratio of factor j to factor i.

$$\boldsymbol{R} = \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1n} \\ r_{21} & r_{22} & \cdots & r_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ r_{n1} & r_{n2} & \cdots & r_{nn} \end{bmatrix} \tag{8}$$

The coherence of human thinking and judgment is reflected in the consistency of a fuzzy judgment matrix, which plays a crucial role in constructing an accurate fuzzy judgment matrix. Due to the intricate nature of comprehensive diagnosis for LS-HSS health, the constructed judgment matrix often lacks internal consistency. Therefore, it is imperative to transform the fuzzy judgment matrix into a coherent and consistent form by following these steps:

Calculate row sums: determine the sum of each row in the established judgment matrix, as shown below:

$$p_i = \sum_{j=1}^{n} r_{ij} (i, j = 1, 2, \ldots, n) \tag{9}$$

Establishing a prioritization relationship matrix,

$$\boldsymbol{P} = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1n} \\ p_{21} & p_{22} & \cdots & p_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ p_{n1} & p_{n2} & \cdots & p_{nn} \end{bmatrix} \tag{10}$$

Where $p_{ij} = \left(p_i - p_j\right)/2n + 0.5$, ($i$, j = 1, 2, n).

Then, the eigenvectors of the fuzzy consistent judgment matrix should be calculated separately using the single-layer weight formula (11), which represents the relative weights of various criteria and sub-criteria.

$$w_i = \frac{1}{n} - \frac{1}{2a} + \frac{1}{na}\sum_{k=1}^{n} r_{ik}, (i = 1, 2, ..., n) \tag{11}$$

Where n represents the order of R, $a = (n-1)/2$.

(4) The fuzzy evaluation membership functions are established, and the membership degrees are calculated to conduct a comprehensive health assessment. The membership degree reflects the extent of belongingness of each element to the fuzzy set of indicator levels. A higher membership degree indicates a stronger belongingness to that indicator level fuzzy set. This study constructs membership functions based on a semi-trapezoidal step method, following the formula for the membership function [77].

$$U_{vA} = \begin{cases} 1, x \le x_1 \\ \dfrac{x_2 - x}{x_2 - x_1}, x_1 < x < x_2 \\ 0, x \ge x_2 \end{cases} \tag{12}$$

$$U_{vB} = \begin{cases} \dfrac{x - a}{x_1 - a}, a \le x < x_1 \\ 1, x_1 \le x < x_2 \\ \dfrac{x_3 - x}{x_3 - x_2}, x_2 \le x < x_3 \\ 0, x \ge x_3 \end{cases} \tag{13}$$

$$U_{vC} = \begin{cases} 0, x \le x_1 \\ \dfrac{x - x_1}{x_2 - x_1}, x_1 < x \le x_2 \\ 1, x_2 < x \le x_3 \\ \dfrac{b - x}{b - x_3}, x_3 < x \le b \end{cases} \tag{14}$$

$$U_{vD} = \begin{cases} 0, x \le x_2 \\ \dfrac{x - x_2}{x_3 - x_2}, x_2 < x \le x_3 \\ 1, x \ge x_3 \end{cases} \tag{15}$$

Where x is the diagnostic indicator value, while the evaluation standard values for grades A, B, C, and D are represented by x_1、x_2、x_3 respectively, the inflection points on the abscissa of the trapezoidal membership function a and b can be determined by calculating the average value and standard deviation of sample data while considering the characteristics of a standard distribution curve. Alternatively, these points can be manually adjusted to ensure comprehensive coverage of all sample data. By substituting each indicator data into the membership function, we can calculate their respective degrees of membership in relation to criteria and form a corresponding matrix $\boldsymbol{J}$ [78].

$$\boldsymbol{J} = \begin{bmatrix} j_{11} & j_{12} & j_{13} & j_{14} \\ j_{21} & j_{22} & j_{23} & j_{24} \\ \vdots & \vdots & \vdots & \vdots \\ j_{n1} & j_{n2} & j_{n3} & j_{n4} \end{bmatrix} \tag{16}$$

The comprehensive health assessment vector $\boldsymbol{A}$ of LS-HSS can be represented as follows [78]:

$$\boldsymbol{A} = \boldsymbol{RWJ} \tag{17}$$

Where $\boldsymbol{R}$ represents the weight coefficient matrix of the criteria layer, $\boldsymbol{W}$ represents the weight coefficient matrix of specific indicators under each criterion layer, and $\boldsymbol{J}$ represents the membership degree matrix of various indicator rating standards. Based on the maximum membership degree index of the overall objective diagnostic value, the LS-HSS health grade assessment is finally determined.

4.4. Knowledge databases of LS-HSS intelligent operation and maintenance platform

In the current landscape characterized by insufficient research on the fault mechanisms of LS-HSS, a scarcity of reference knowledge and expert resources, challenges in the summary and reuse of knowledge and experiences, and limited universality, the digital twin aims to achieve virtual-real interactive collaboration. It utilizes DT data to continuously enhance the optimization of LS-HSS maintenance processes, enriching knowledge data related to the operation and maintenance of LS-HSS. Establishing a knowledge database is crucial for implementing intelligent decision-making and dynamic optimization strategies to further enhance the level of intelligence in applying knowledge to the operation and maintenance of LS-HSS.

Firstly, the knowledge ontology is constructed based on structured files (e.g., a simulation database) within the LS-HSS management system. Simultaneously, unstructured and semi-structured knowledge transforms structured knowledge through extraction and processing, including rule files, standard files, and maintenance manuals. Subsequently, description information of the same entity or relationship is merged before storing and continuously updating the knowledge during application. The intelligent operation and maintenance knowledge graph of LS-HSS is illustrated in Fig. 14.

Fig. 15 illustrates the knowledge-driven intelligent decision-making and dynamic optimization strategies for LS-HSS. This process involves four steps: dynamic assessment, issue recognition, root cause analysis,

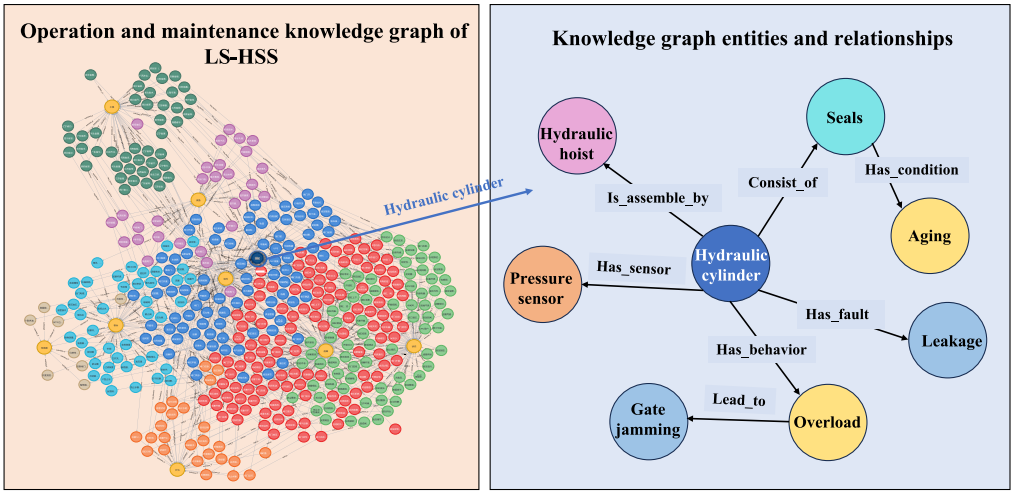


Fig. 14. Intelligent operation and maintenance knowledge graph of LS-HSS (Part).

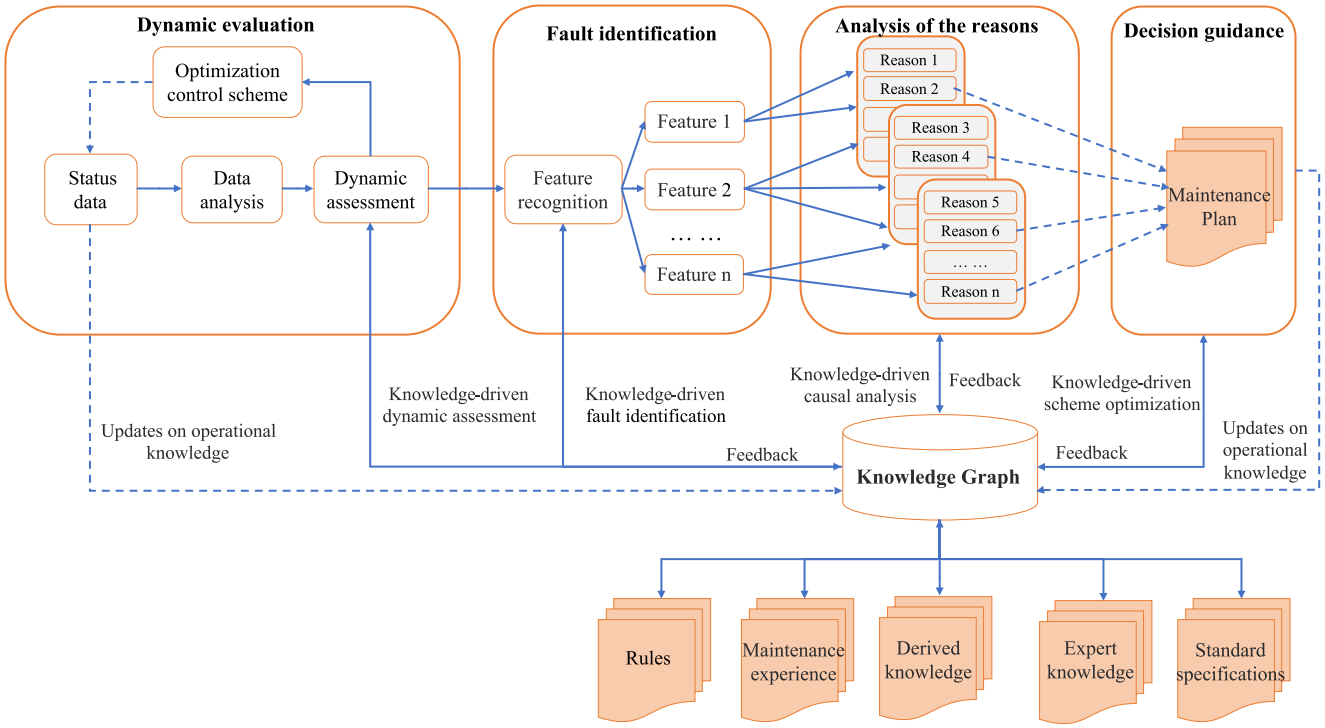


Fig. 15. The flowchart of the dynamic optimization process for intelligent decision-making.

and decision guidance. An operation and maintenance knowledge base is essential to achieve intelligent generation and dynamic optimization of operation and maintenance plans for LS-HSS under various scenarios and complex conditions. This requires integrating industry-specific data, such as historical maintenance plans, business rules, expert knowledge, and industry standards. Knowledge graph technology is central to this process, encompassing knowledge data preprocessing, ontology construction, knowledge generation, and knowledge refinement. With a focus on the operation and maintenance of LS-HSS, the approach involves the establishment of a knowledge-driven framework for real-time dynamic assessment of operational status, automatic identification of abnormal characteristics, autonomous diagnosis of failures, and the generation of emergency response plans. This closed-loop control capability enhances the accuracy and practicality of maintenance plans at different stages of the operation and maintenance process, leading to a continuous enhancement in intelligent operation and maintenance capabilities throughout the entire service life of LS-HSS.

5. Implementation and evaluation

The DT-IOM platform has been constructed using the DT-IOM framework and essential technologies. This platform has demonstrated successful implementation in Luhun Reservoir, which is recognized as one of Luoyang City's top three reservoirs. The construction of a digital twin model for the LS-HSS of the spillway aims to replicate and verify the complete operational procedure of the system, offering an opportunity for practical application and showcasing its demonstrative capabilities, as shown in Fig. 16.

5.1. 3D modeling and updating services of LS-HSS

The working condition of the LS-HSS is illustrated in Fig. 17 (a), with a width of 12.0 m and a height of 10.5 m. To guarantee the authenticity of the 3D virtual representation of the actual physical entity, the physical entities are modeled using 3D Max software (Autodesk, USA). Fig. 17 (b) shows the 3D virtual model of the whole spillway of the reservoir. Fig. 17 (c) shows the 3D virtual model of the LS-HSS. The portion of the 3D virtual model is depicted in Fig. 17 (d)-(e).

Furthermore, to ensure consistency between the virtual model and the physical entity when changes are made due to LS-HSS maintenance and replacement, dynamic updates are achieved by uploading the virtual model through the system's model maintenance and update management functions, as depicted in Fig. 18.

5.2. Digital twin-driven monitoring services (Forecasts and issue early warnings)

During the operation of LS-HSS, equipment performance may deteriorate, and safety risks can increase due to changes in external environmental conditions, material property degradation, and inadequate maintenance. Therefore, accurately detecting anomalies, analyzing their underlying causes, conducting thorough fault investigations, and promptly resolving issues are paramount in preventing significant failures.

The real-time mapping and online monitoring system facilitates the digital twin-driven monitoring service for LS-HSS. The system can integrate models, data, and algorithms for real-time operational status assessment and anomaly diagnosis.

The real-time mapping and monitoring system developed by the Unity 3D engine, as depicted in Fig. 19, encompasses various functionalities such as real-time monitoring information, security status information (e.g., anomalies early warning information and fault diagnosis information), system configuration information, and manual inspection information.

The real-time status information of crucial components for LS-HSS, including the radial gate section, hydraulic hoist section, and electrical control system section, is depicted in Fig. 20. It can provide remote, live monitoring. This guarantees transparent equipment status data, understandable operational dependability, and prompt notification of unusual circumstances. For example, when abnormal conditions such as stress, vibration, or posture occur during the radial gate section's opening and closing process, the monitoring system's real-time status information will be updated to display a fault alarm and notify the user to handle the issue. This ensures equipment operates normally and

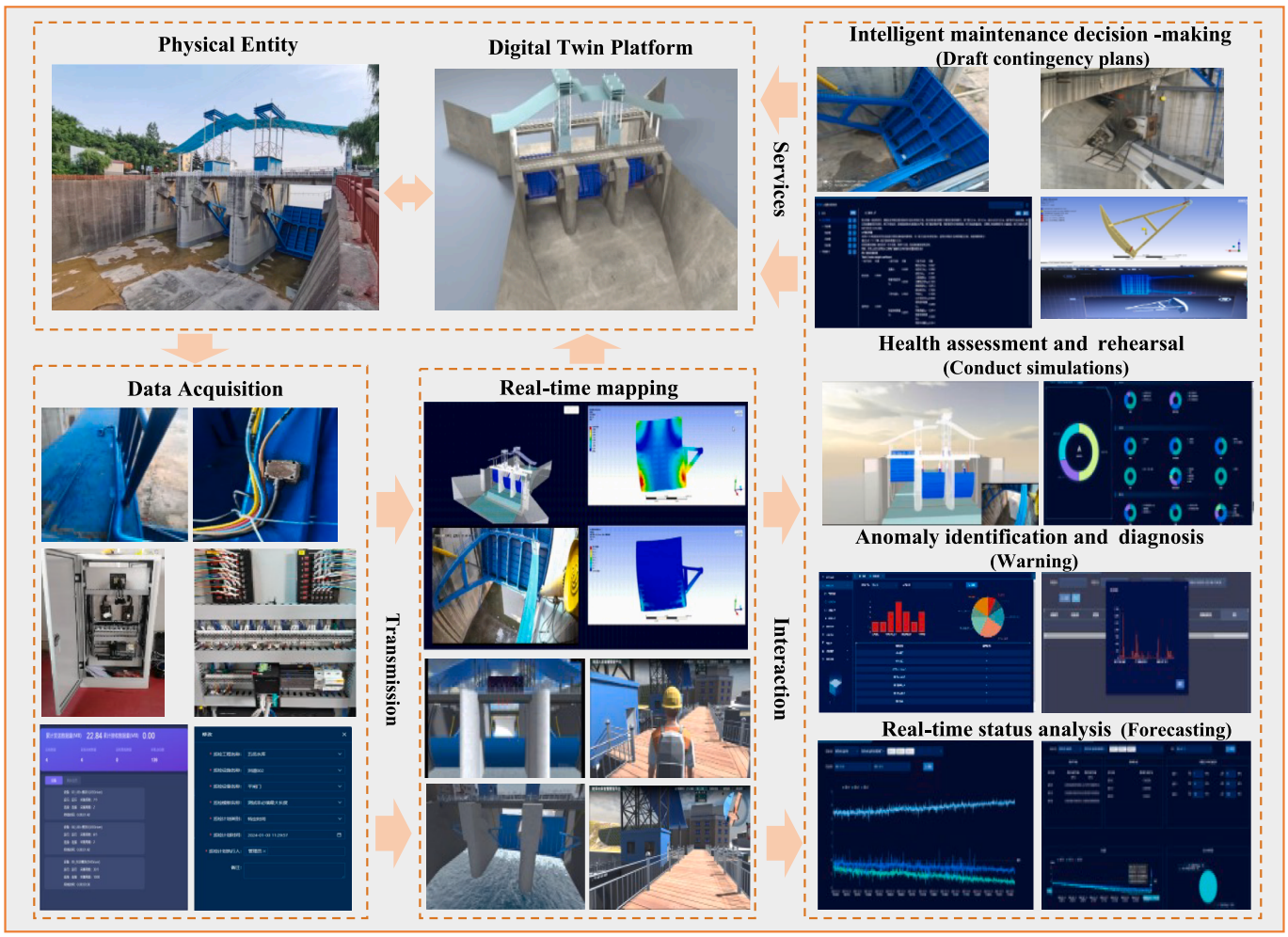


Fig. 16. Case study of intelligent operation and maintenance of a spillway.

stably during water resource scheduling by adjusting control parameters and performing necessary maintenance.

The detailed flowchart for digital twin-driven monitoring services for LS-HSS is shown in Fig. 21. Following is an overview of the process: First, establish a fault-code system. Whenever an abnormal state is detected during operation, record the circumstances of each occurrence. Simultaneously, link the corresponding operating condition data (such as control parameters, operational data, environmental parameters, recent maintenance data, etc., related to *MPI/WPI/OMI*) with each recorded abnormal state. This process constructs a dataset of abnormal operational conditions. Next, a comparative analysis using a historical fault feature knowledge base will be conducted. Record the operating condition data if the abnormal state matches the historical fault feature knowledge base. Then, the maintenance method knowledge base is used to formulate a comprehensive plan for diagnosis and maintenance. If fault features are not correctly identified, conduct a synchronous diagnostic analysis of the fault in physical and virtual spaces based on the abnormal operational condition feature dataset. This is achieved using mechanism-driven simulation models (statics simulation models, dynamics simulation models, etc.) and data-driven fault diagnosis models (decision trees, support vector machines, neural networks, etc.). The obtained diagnostic results are subsequently subjected to consistency checks based on expert judgment. In cases of a significant deviation between the virtual-real diagnostic results, seek expert remote consultations to further enhance the accuracy and reliability. Finally, based on the diagnostic results, determine the fault location, clarify the cause, and formulate a maintenance plan. This information is shared with users, which may involve generating maintenance work orders, log information, etc. The fault diagnostic information, including fault features and solutions, should be utilized to update the knowledge in the operational condition database, fault feature knowledge base, and maintenance method knowledge base simultaneously.

5.3. Online assessment and rehearsal of LS-HSS operating conditions (Conduct simulations)

Health assessments throughout the lifecycle are crucial in ensuring the reliability and safe operation of LS-HSS, considering their impact on equipment structures and performance during abnormal conditions, particularly maintenance activities. The DT-IOM platform enables functions such as the online health assessment (as shown in Fig. 22), simulation of operation status (as shown in Fig. 23), and prediction of future status (as shown in Fig. 24). The online assessment and rehearsal model (health assessment model, finite element model, or surrogate model) utilizes real-time status data, manual inspection data, and simulation data to accomplish health status evaluation, which keeps in line with the current operating status or predicts the health status of LS-HSS under hypothetical operating conditions.

5.4. Intelligent maintenance decision-making services (Draft contingency plans)

The DT-IOM platform's knowledge databases have been utilized to develop an intelligent question-answering system based on a knowledge

Fig. 17. (a) Working condition of the LS-HSS, (b) 3D virtual model of the hydraulic structures, (c) 3D virtual model of the LS-HSS, (d) hydraulic station and electrical control system section, (e) radial gate section.

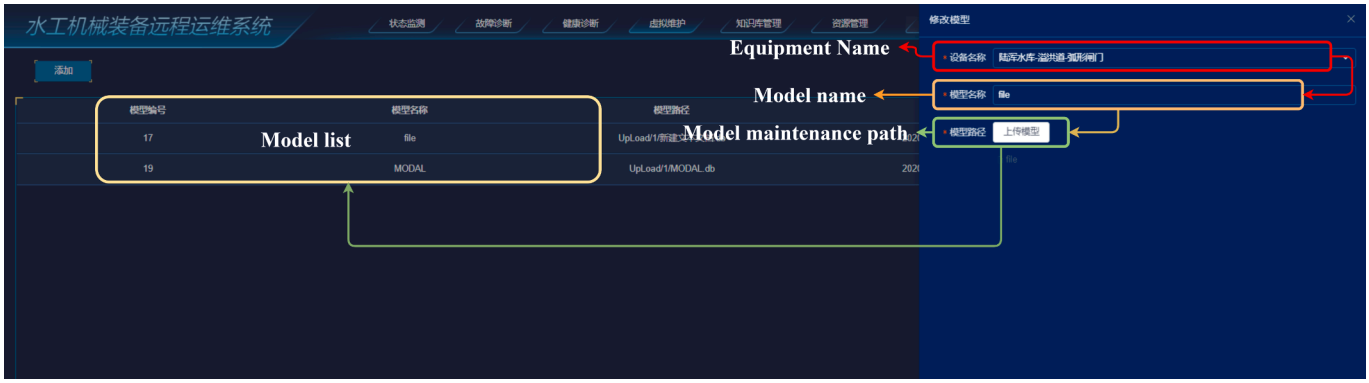


Fig. 18. Model maintenance and update management.

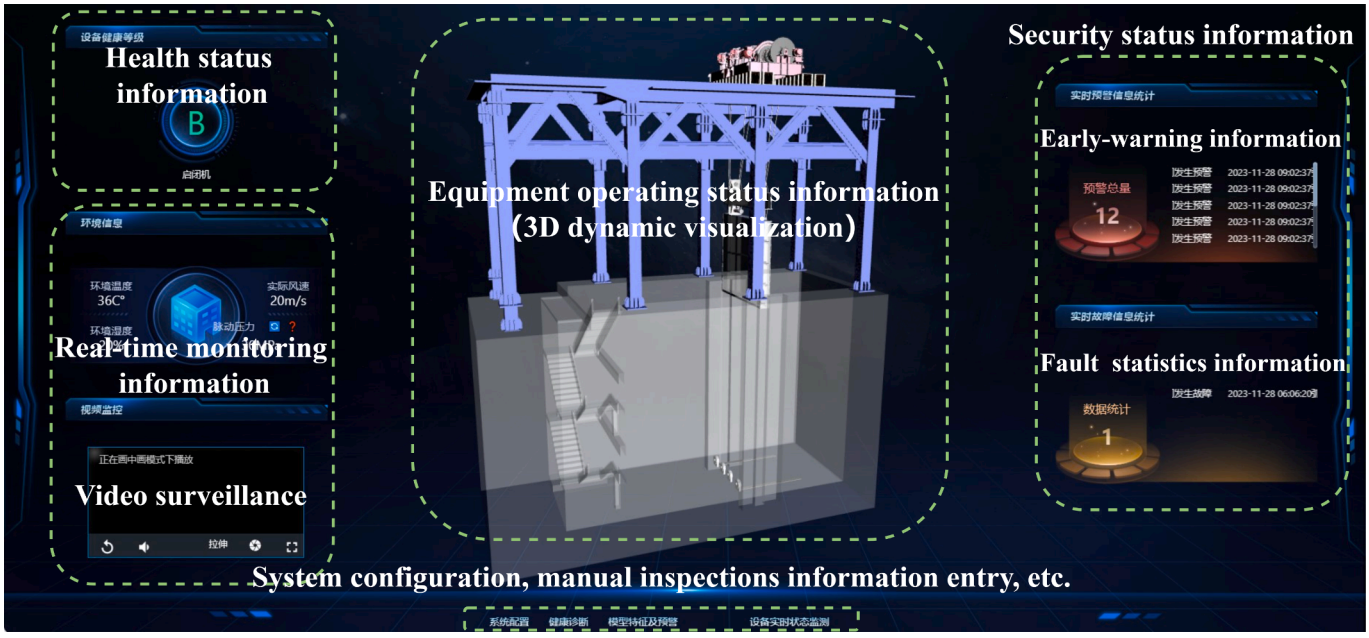


Fig. 19. Digital twin-driven real-time mapping and monitoring system of LS-HSS intelligent operation and maintenance platform.

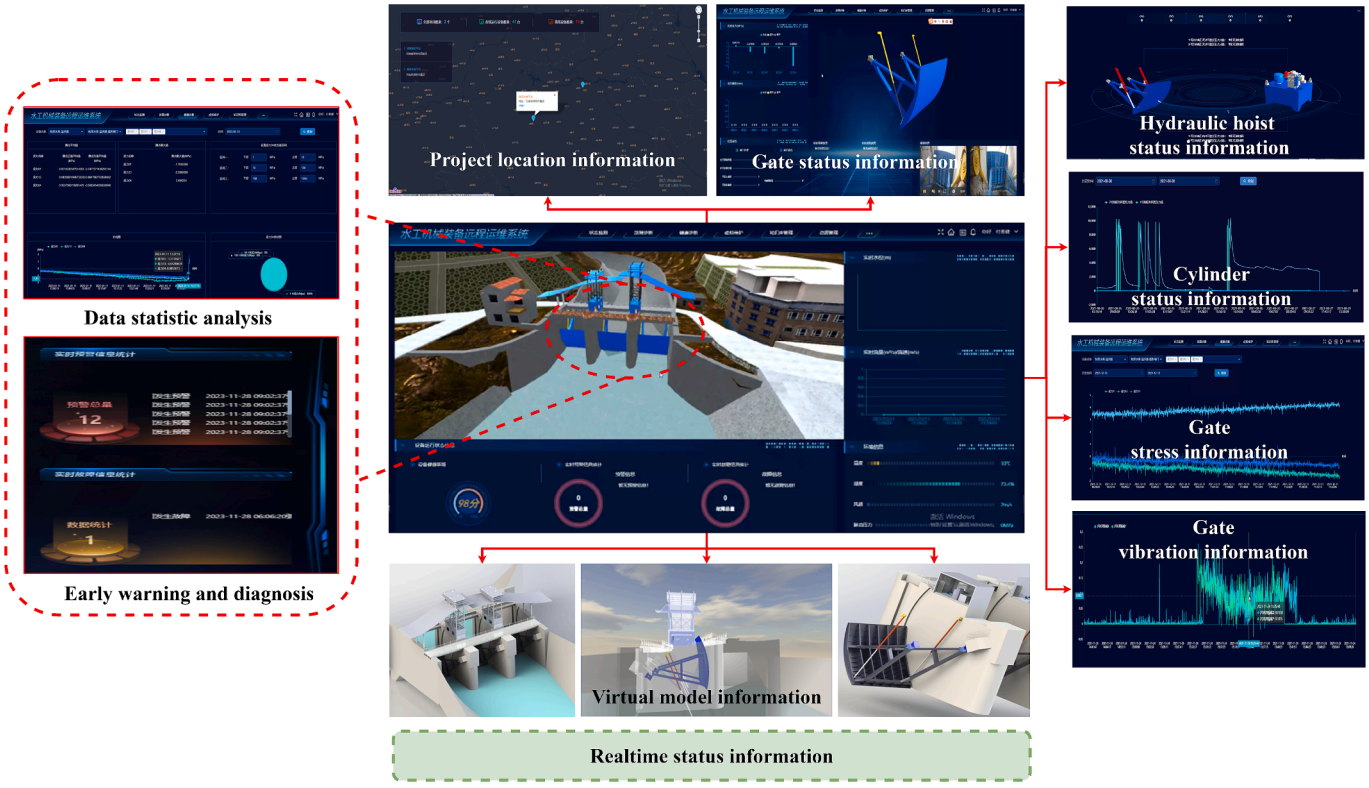


Fig. 20. Real-time status information of crucial components for the DT-IOM platform.

graph. Intelligent knowledge management and application for hydraulic actuator faults are successfully achieved by integrating user interaction and semantic queries.

The DT-IOM platform's knowledge databases construct an intelligent question-and-answer system based on a knowledge graph, effectively integrating user interaction and semantic query. This facilitates the intelligent management and application of maintenance data.

The framework of the intelligent question-answering system, as depicted in Fig. 25 based on the knowledge graph, comprises several components as follows:

(1) Natural language query: accepting the user's natural language questions through the interactive interface.
(2) Query classification: identifying the dialogue from user input.
(3) Query parsing: utilizing pre-trained models to comprehend the user's intent and context, including the jieba word segmentation

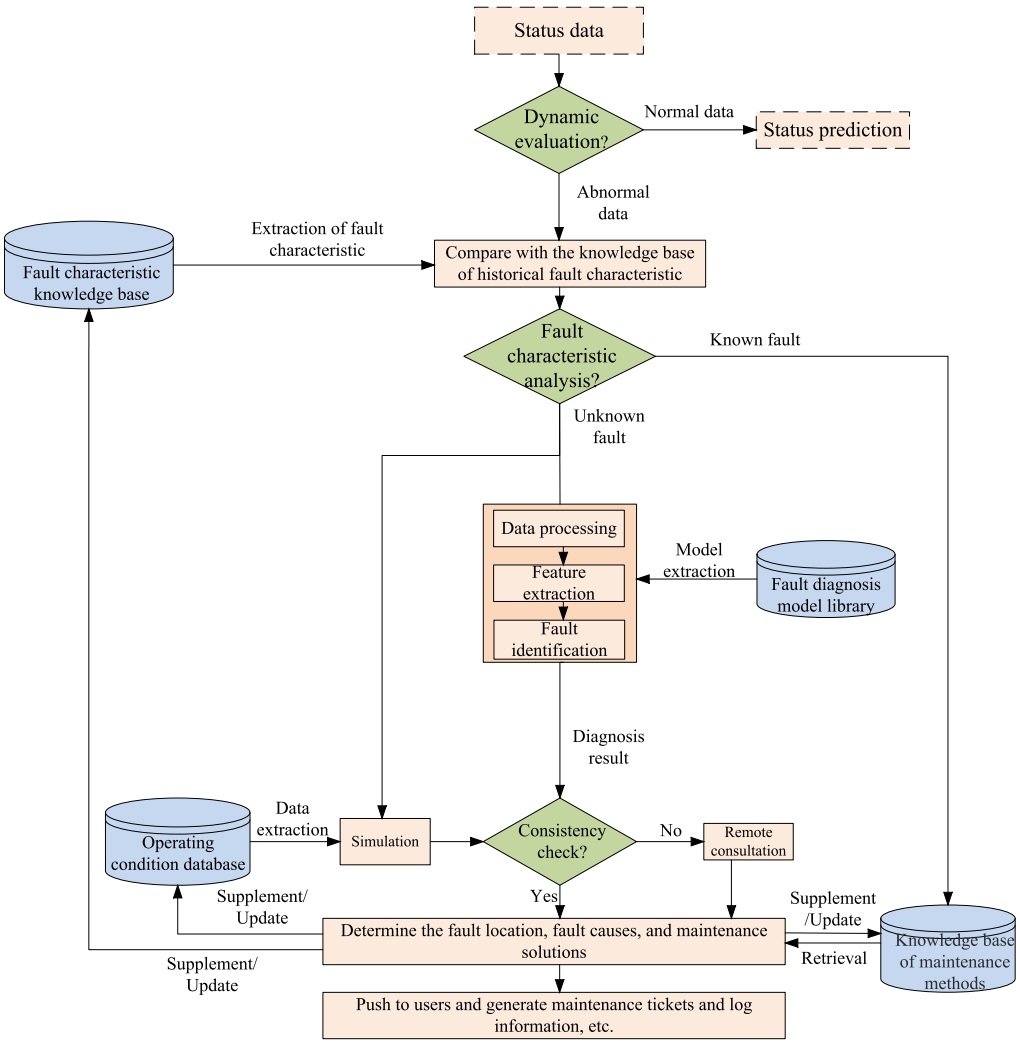


Fig. 21. The flowchart for digital twin-driven monitoring services.

tool [79], BERT-Bi-LSTM-CRF entity recognition model, and BERT-Bi-LSTM-Attention intent recognition model [80].

(4) Query transformation: constructing a query expression using predefined question templates to query the knowledge base.
(5) Result retrieval: retrieving information from the graph database based on determined question categories, providing users with desired answers, and returning them to the interactive interface.

The functional interface of the question-answering system, as depicted in Fig. 26, demonstrates intelligent maintenance decision-making through interactive questioning using natural language. This system is built on Flask and Vue frameworks.

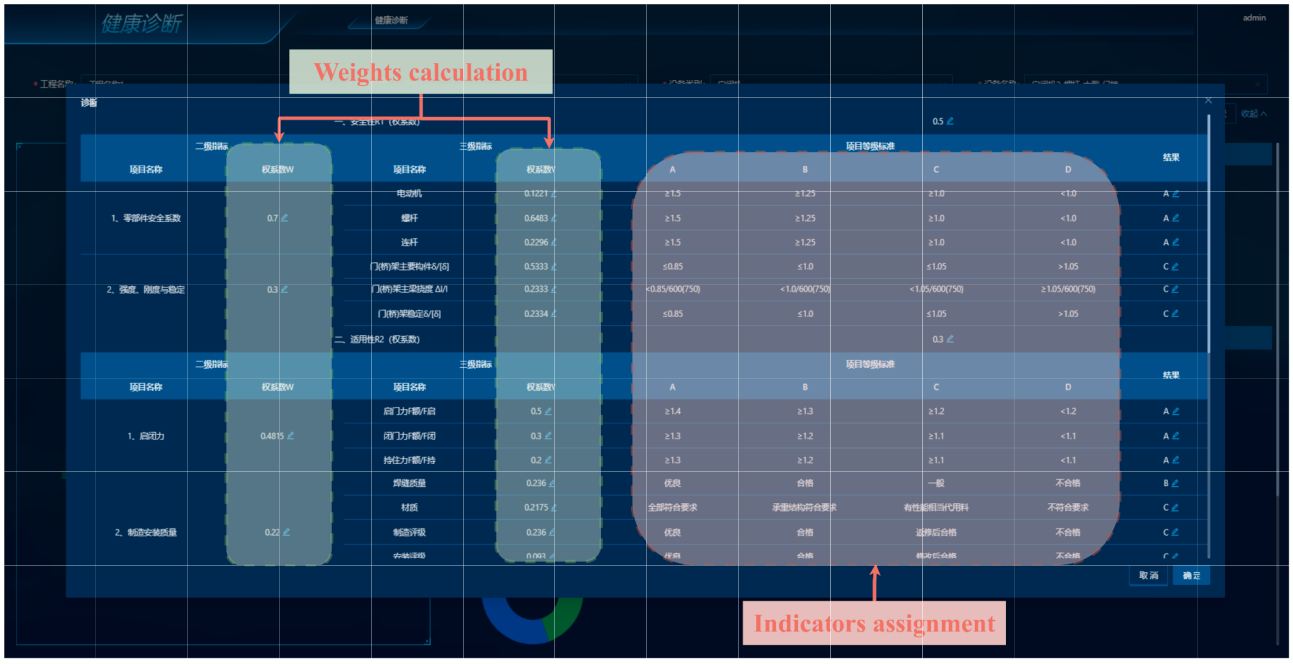
5.5. *Performance of LS-HSS intelligent operation and maintenance platform*

Through the deployment and application at the Luhun Reservoir, this DT-IOM platform effectively enhances the digital identification and diagnosis of performance evolution characteristics of LS-HSS during long-term service and maintenance levels. It plays a crucial role in ensuring the safe and reliable operation. Firstly, based on the response surface methodology, Table 3 compares vital parameters before and after structural optimization. These parameters encompass central structural thickness (e.g., main beam, stiffener, radial arm, panel, main girder), Poisson's ratio, elasticity modulus (N/m^2), and density (kg/m^3).

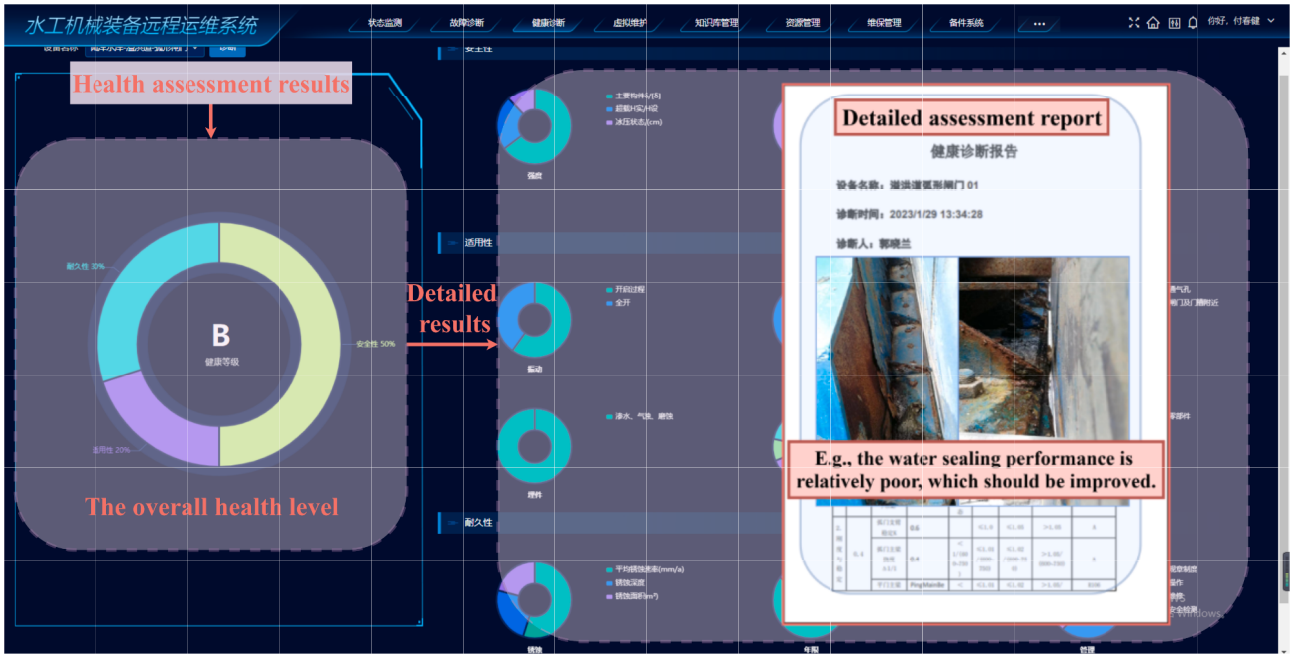
Conduct simulation analysis on the finite element model based on optimized parameters. The comparative analysis results, as shown in Table 4 below, demonstrate that the optimized parameters applied to the finite element model significantly reduced errors by 2.78 %, 3.57 %, 4.63 %, 7.55 %, 1.74 %, and 7.13 %, respectively, when compared to the initial experience-driven simulation results (Empirical Parameters), with online monitoring values serving as a reference.

The diagram in Fig. 27 illustrates the effective identification of LS-HSS anomalies and adjustment of control parameters to mitigate abnormal vibrations during the radial gate opening and closing process based on wavelet packet information entropy and improved SVM [81], ensuring safe operation through online monitoring and real-time data analysis. Additionally, Fig. 28 showcases a comprehensive evaluation model [75] for assessing the health of LS-HSS by integrating online monitoring and inspection data. By leveraging fault knowledge graphs, targeted maintenance can be performed on potential problem components to enhance equipment safety.

Overall, digital twin-driven intelligent operation and maintenance for LS-HSS employs advanced techniques, including data service, real-time mapping, online monitoring, health assessment, and intelligent decision-making methods. These key technologies collectively enable dynamic virtual-real interaction and establish a closed-loop proactive safety assurance and predictive maintenance mode with the stages of perception, analysis, decision-making, and optimization, as illustrated in Fig. 29.



(a)



(b)

Fig. 22. Online health assessment of LS-HSS, (a) gives weights to indexes by FAHP and assignment [75], (b) calculates the health level [75].

The DT-IOM platform has been successfully implemented in multiple hydraulic engineering projects, such as Luhun Reservoir, Heiquan Reservoir, Hekoucun Reservoir, and Wuyue Pumped Storage Power Station. Fig. 30 illustrates the platform's functionalities and applications.

At present, the platform is primarily utilized for IoT data collection, virtual-real interaction, and 2D/3D visualization functions, catering to the intelligent operation and maintenance requirements of LS-HSS such as radial gates of reservoir spillways, plain gates of flood discharge orifices and emergency gates of pumped storage power stations. Taking the Luhun Reservoir as an example, compared with conventional manual inspections and experience-based maintenance (as illustrated in Table 5), the digital twin platform allows for real-time monitoring of equipment operation status and abnormal situations, enabling more timely management and response. The platform has significantly enhanced inspection efficiency and accuracy through online monitoring and VR-based intelligent inspections. Simultaneously, it has reduced labor intensity while effectively improving the digital recognition capabilities of performance evolution characteristics for LS-HSS during long-term service. The average equipment inspection time has been reduced by approximately 50 %. Moreover, when there is an abnormal situation, the inspection personnel can directly initiate a maintenance request in the application. The system will automatically notify the maintenance engineer and record detailed information about the

Fig. 23. Simulation of online running status based on finite element model.

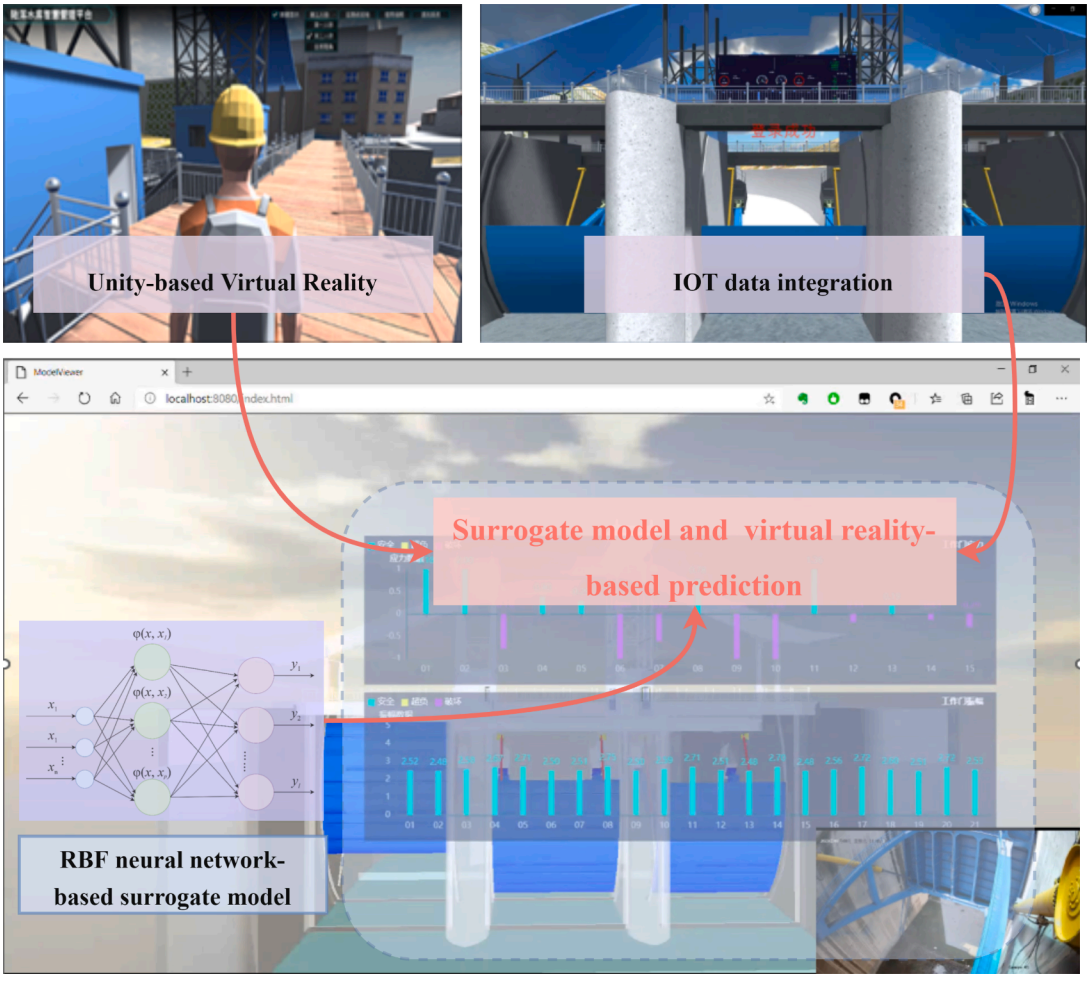


Fig. 24. Prediction of future status through the surrogate model and virtual reality.

abnormal situation, achieving quick response and handling. The operation and maintenance response and handling efficiency is improved by over 60 %. In addition, by conducting dynamic health assessments and multi-operational simulations, the equipment's health status is effectively evaluated, leading to proactive maintenance. Consequently, the equipment failure rate has been reduced by over 20 %, enhancing operations' safety and reliability. China's Ministry of Water Resources Science and Technology Promotion Center has incorporated these technological achievements into the Key Promotion and Guidance Catalogue for Advanced Practical Technologies (TZ2021277).

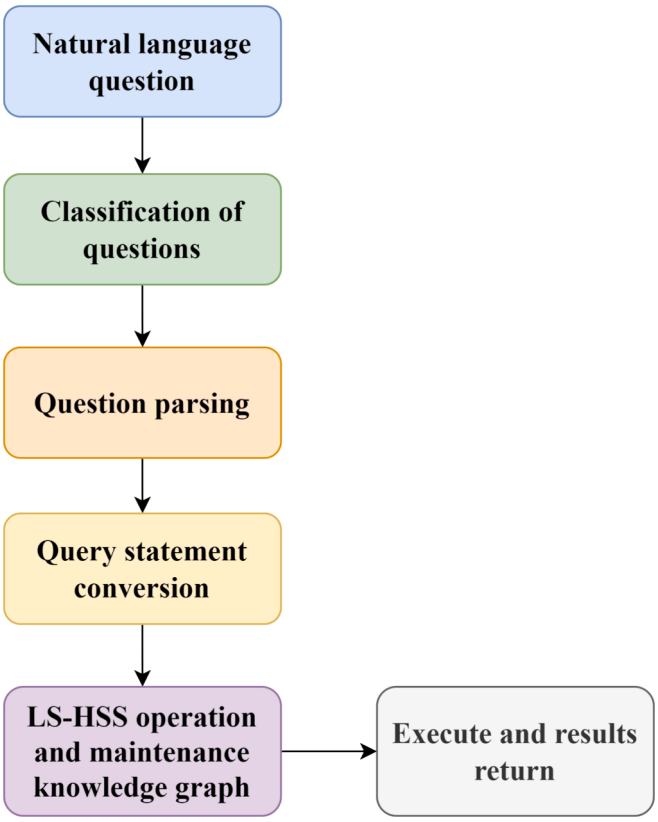


Fig. 25. Framework of the question-answering system based on knowledge graph.

The DT-IOM platform will play a comprehensive role in enhancing informatization in LS-HSS operation and maintenance processes as we enter the next stage, characterized by continuous intelligentization in hydraulic engineering and implementing digital twin basin projects. By profoundly integrating engineering data, models, knowledge, and services, it will continue to drive the digital and intelligent transformation of LS-HSS operation and maintenance processes.

6. Discussions

In the conventional operation and maintenance process, common challenges arise due to the extensive size and intricate operational conditions of LS-HSS, as well as the manifold influences from natural degradation and variations in complex water conditions during its prolonged continuous service, as follows: a) The harsh operating conditions and environmental factors on-site make it challenging to conduct inspections with efficiency and accuracy; b) Natural degradation and fluctuating water conditions give rise to diverse forms of failures, making it difficult to conduct early warning and accurate diagnosis; c) It is challenging to identify safety risks and assess the health status, which impedes the implementation of predictive maintenance.

Earlier studies have explored the application and effects of the digital twin in hydraulic engineering, primarily focusing on areas such as dam safety monitoring, hydroelectric generating units, and watershed water resource management, as shown in Table 6. In comparison to conventional digital technologies, the digital twin necessitates enhanced

Table 3
Comparison of main parameters before and after structural optimization.

Parameters	Initial design parameter values	Modified parameter values
Thickness of main beam (mm)	16	14.536
Thickness of stiffener (mm)	20	19.849
Thickness of radial arm (mm)	16	16.129
Thickness of panel (mm)	12	12.987
Thickness of main girder (mm)	12	12.544
Poisson's ratio	0.3	0.288
Elasticity modulus (N/m^2)	2.06E11	1.94E11
Density (kg/m^3)	7850	7784.37

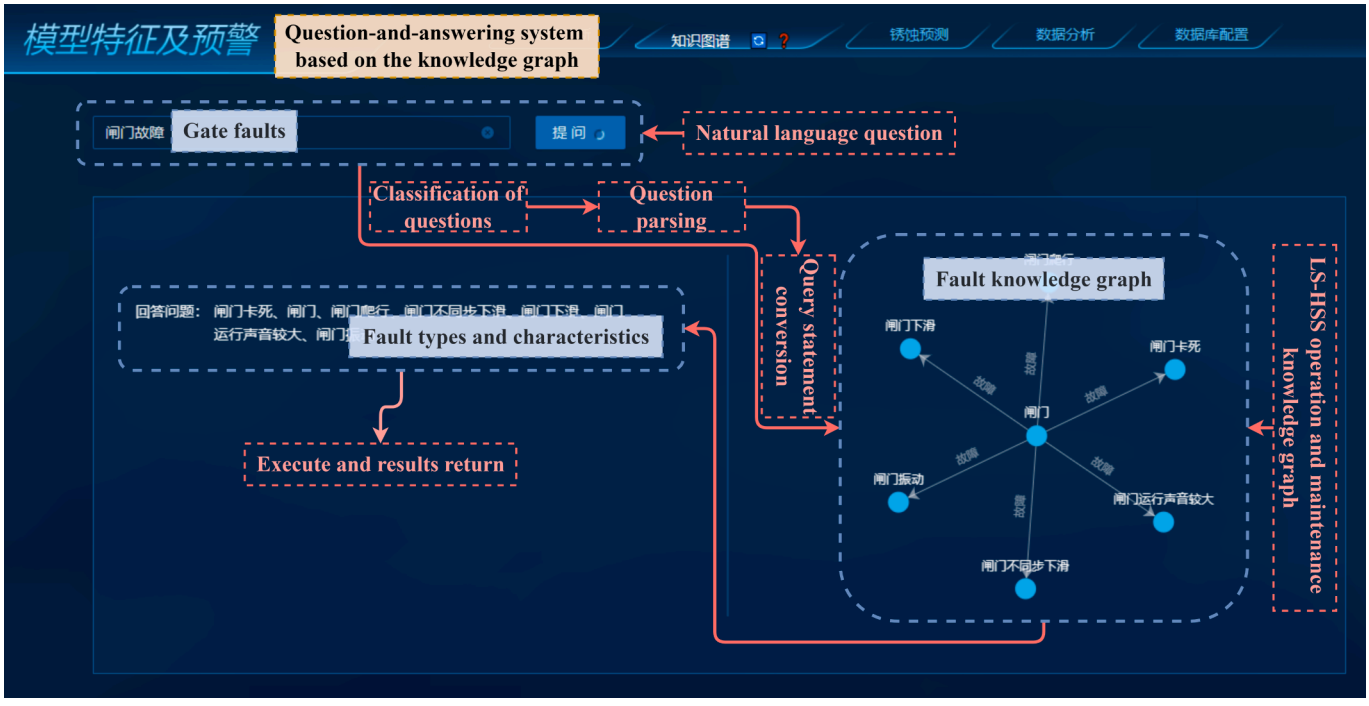


Fig. 26. Interface of the primary function of the question-and-answering system based on the knowledge graph.

Table 4
Analysis of the main parameters optimization results of the finite element model.

Parameter	First-order frequency (Hz)	Second-order frequency (Hz)	Main beam stress (MPa)	Arm stress (MPa)	Panel stress (MPa)	Main longitudinal beam stress (MPa)
Measured results	3.6	12.5	2.8874	0.8971	0.4951	0.4073
Empirical results	3.7778	12.96	2.4112	1.2544	0.6161	0.3283
Optimization results	3.6731	12.497	2.5228	1.1597	0.6054	0.3517
Variation value	+0.1047	+0.003	−0.1116	+0.0947	+0.0107	−0.0234
Optimization rate	+2.78 %	+3.57 %	+4.63 %	+7.55 %	+1.74 %	+7.13 %

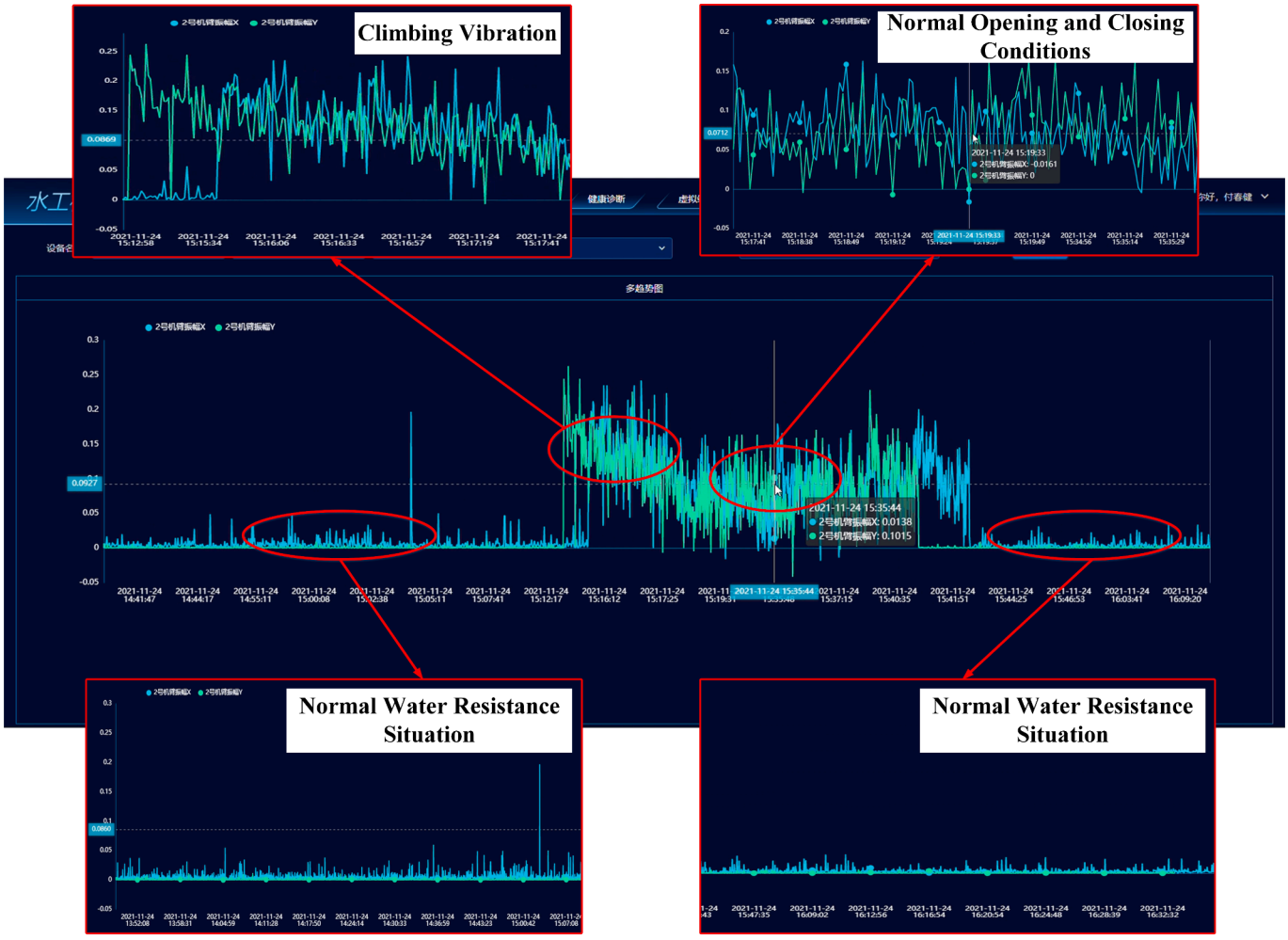


Fig. 27. Real-time data of vibration parameters during the opening and closing process.

capabilities such as seamless integration of virtual and physical components, real-time interaction, iterative optimization, and a data-driven approach to function as a unified entity that encompasses both the tangible object and its corresponding digital representation, enabling seamless interconnectivity, fostering collaboration, and co-evolving in tandem, as shown in Fig. 31.

However, as illustrated in Table 1, the primary focus of DT-based approaches in LS-HSS lies in monitoring, information modeling, multi-physics modeling, fusion of simulations, or dynamic model adjustment. Insufficient attention has been given to the impact of digital twin technology on enhancing the performance of LS-HSS intelligent maintenance management and corresponding digital twin platform research. By analyzing the current representative research findings on the fundamental theory and implementation approach of digital twin (as presented in Table 7, a solid theoretical and methodological foundation has been established for enabling intelligent operation and maintenance of LS-HSS. Nevertheless, due to the intricate service conditions and environmental factors, applying relevant technical theories in practical industrial applications without adjustment and improvement is not suitable and feasible. It becomes imperative to consider differentiated demands in data collection, analysis methods, integration of knowledge and models, and service encapsulation. Therefore, it is essential to establish a comprehensive framework for digital twin modeling in LS-HSS operation and maintenance. This includes integrating data, models, knowledge, and services driven by digital twin technology. Additionally, establishing a comprehensive DT-driven LS-HSS intelligent operation and maintenance platform becomes crucial in facilitating the digital transformation of conventional LS-HSS operations.

Hence, in this study, a five-dimensional digital twin modeling framework for LS-HSS operation and maintenance is proposed, with a

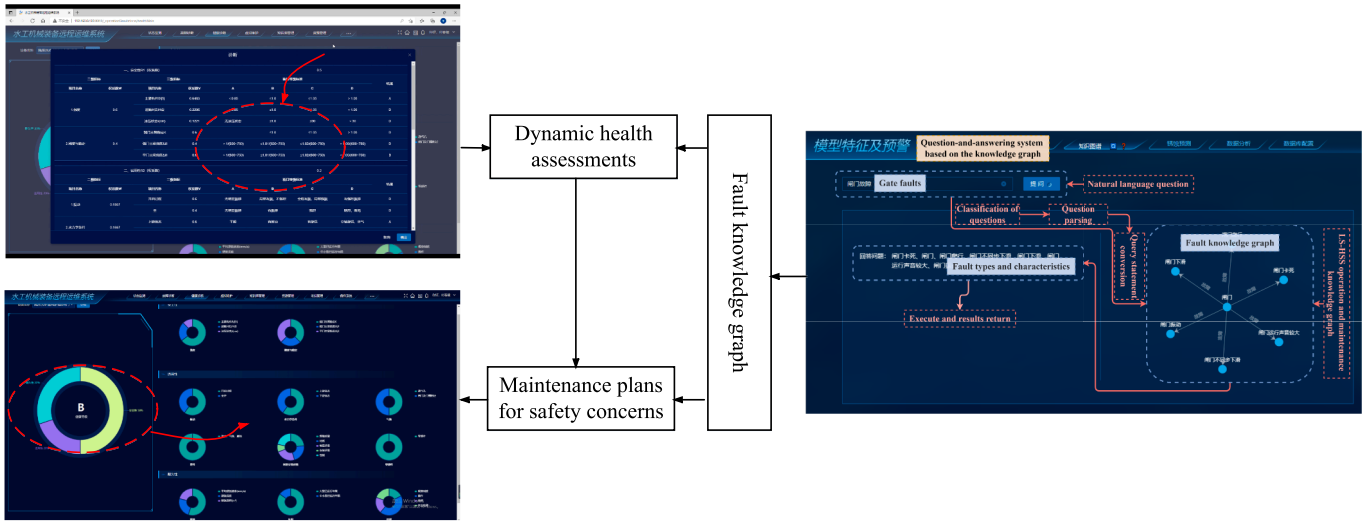


Fig. 28. Maintenance of crucial components based on health assessment results and knowledge graph.

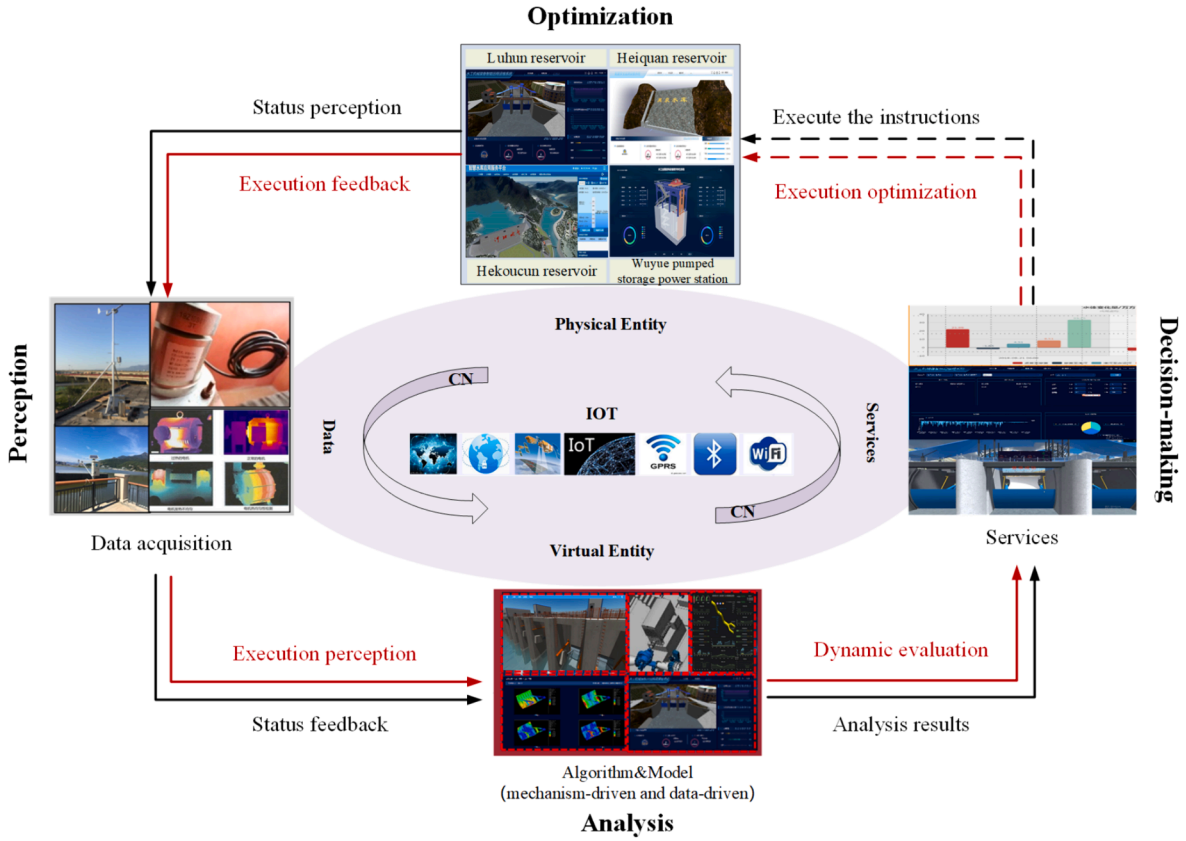


Fig. 29. A closed-loop intelligent operation and maintenance process for LS-HSS based on digital twin.

specific focus on the digital transformation pathway of conventional operation and maintenance, as well as the integration methods of data, models, knowledge, and services driven by digital twin technology. The DT-IOM platform for LS-HSS, validated through practical engineering applications, highlights the following functional advantages: capability of IoT connection, modeling and integration, virtual-real interaction, 2D/3D visualization, and service expansion, as depicted in Fig. 32.

(1) IoT connection capability. Facilitate the real-time data collection and integration of physical entities or systems, such as vibration, stress, attitude, operating status, and environmental parameters, utilizing PLC, sensors, and dynamic data acquisition instruments.
(2) Modeling and integration capabilities. The DT-IOM platform utilizes essential components of LS-HSS, environmental conditions, and hydraulic characteristics as modeling entities to construct a digital model. It also organizes data, incorporates analytical algorithms, and forms an integrated digital twin model that combines data, model, and computation. This facilitates subsequent utilization in various digital twin application scenarios for knowledge sharing and reuse.

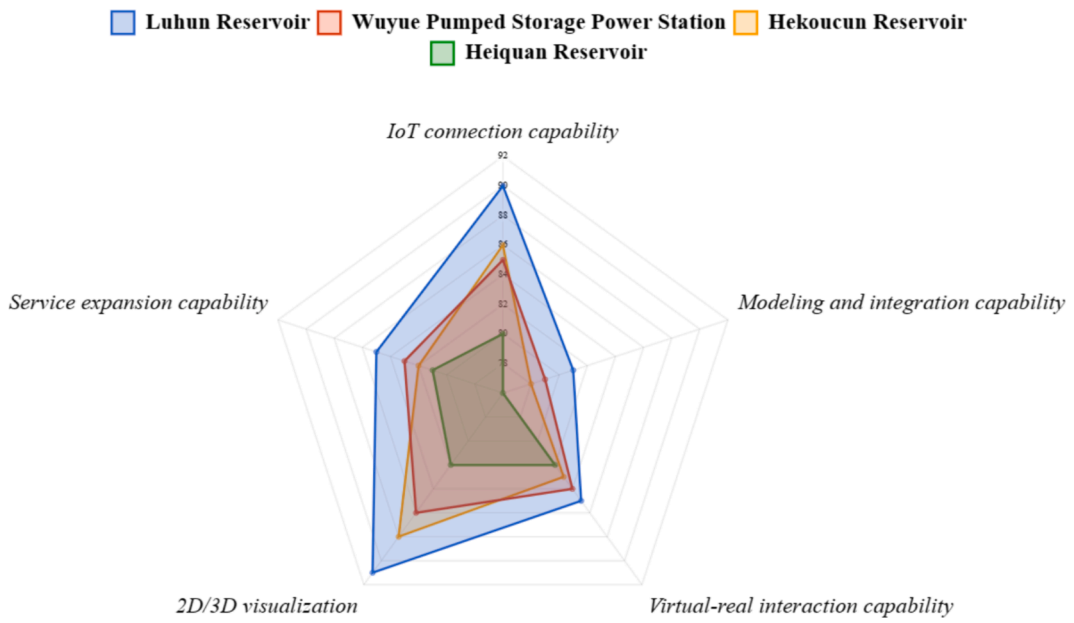


Fig. 30. Functionalities and applications of the DT-IOM platform in hydraulic engineering.

Table 5
Comparison of conventional operation and maintenance approach (taking the spillway's LS-HSS as an example). *Provided by Luhun Reservoir operation management center.*

Categories	Routine inspection	Risk recognition and handling response time	Health assessment	Maintenance scheme
Conventional approach	Manual on-site inspection (average 30 min)	1) Manual judgment 2) The average response time is more than 15 min	1) Short-time prototype observation 2) Experts in-situ observation	1) Experience-based judgment 2) Average failure rate 10 %
DT-IOM platform	Online monitoring, virtual reality, and video-based intelligent inspection (average 10 min)	1) The system realizes automatic identification and intelligent analysis 2) The average response time is less than 5 min	1) On-line health assessment 2) Dynamic simulation based on finite element analysis, surrogate model, and virtual reality	1) Data and model hybrid-drive optimization 2) Intelligent decision-making based on the knowledge graph 3) Average failure rate 7.5 %
Performance	The inspection time is reduced by more than 50 %	The efficiency increased by over 60 %	Achieving dynamic evaluation and supporting multiple operating conditions and simulations across various scenarios	The equipment average failure rate has been reduced by over 20 %

(3) Virtual-real interaction capability. The fundamental characteristic of digital twin lies in the integration of virtual and physical entities. Constructing a digital twin model for physical objects can achieve a bidirectional mapping between the virtual model and the physical entity. Users can utilize the platform to attain synchronized interaction and feedback control between the virtual and physical worlds, encompassing action behavior configuration, state synchronization, rule setting, numerical-model association configuration, and consistency verification between the virtual and physical aspects.

(4) 2D/3D Visualization capability. The visualization of multi-source and heterogeneous data enhances the understanding and presentation of the equipment's operational status, thereby enabling interactive visual operations during operation and maintenance processes.

(5) Service expansion capability. By implementing a system-wide data dictionary and utilizing techniques like service encapsulation and composition (e.g., API interfaces or Web Services), it is possible to expose the functionality required for accessing data as dedicated data services. At the same time, embracing a modular design approach that separates front-end from back-end architecture facilitates modularity and service encapsulation of application layer functions. Other systems can then access these functions through API interfaces or Web Services to meet the needs of collaborative construction efforts while enabling seamless sharing of data resources and integration of features within digital twin watershed development.

The digital twin platforms discussed in existing literature (as illustrated in Table 8) are widely used in digital workshops, smart factories, and smart industrial parks/cities construction [82]. Their primary focus is enhancing the universality of complex model construction and integration, the applicability of multiple communication protocols, data standardization, and algorithms' compatibility. Overall analysis reveals that different research fields have varying understandings of digital twin technology systems' value and function. Consequently, platform openness and integration ability differ significantly. Furthermore, different industries prioritize their demand for digital twin platforms differently. Promoting industry digitization, intelligent transformation, and innovation effectively requires deep integration with industry data knowledge and service models. In addition, existing platforms are also limited in terms of applicability, cost-effectiveness, and implementation efficiency. In contrast, the DT-IOM platform proposed in this study combines the technical advantages of the fundamental digital twin platforms, encompassing IoT connection, data processing, modeling/

Table 6
Application and effects of the digital twin in hydraulic engineering.

Application fields	Application object	Key enabling technologies and advantages	References
Hydraulic engineering safety management	Engineering design and construction process	1) BIM (Building Information Modeling) technology is used for three-dimensional collaborative design. 2) It allows different professionals to share data, information, and knowledge, leading to visualization, multi-discipline collaboration, and project optimization.	[9] (2023)
		1) An innovative and efficient framework for motion forecasting in construction sites, utilizing a digital twin-based approach that incorporates cloud-edge collaboration; 2) Key enabling technologies for the proposed edge-cloud collaboration framework include digital twin technology, multi-object tracking, motion forecasting models (such as social LSTM-based models) , and edge computing platform; 3) Enables the prediction of future trajectories of workers while ensuring data privacy and reducing computational burden.	[10] (2024)
		1) Comprehensively investigate the potential of digital transformation in enhancing operational intelligence within the building construction industry; 2) Three primary classifications of crucial digital enabling technologies for digital transformation in the building construction sector have been recognized: data-related technologies, high-fidelity modeling technologies, and model-based simulation technologies.	[11] (2024)
	Engineering operation process	1) A new online sequential extreme learning machine of dam safety monitoring model; 2) Advantages of the AF-OS-ELM-MVE model: real-time prediction, interval prediction, computational efficiency, robustness, and generalization. 3) It provides more effective information for dam safety decision-making, improving the overall understanding of the structures' operational status.	[8] (2024)
		1) Proposes an idea to apply the digital twin technique to hydraulic engineering operation management; 2) Enhances the level of hydraulic engineering operation management and provides valuable insights for further development of China's hydraulic engineering.	[12] (2018)
		1) Describes the reservoir safety monitoring system in the context of intelligent water conservancy, focusing on intelligent perception, infrastructure, data resources, intelligent analysis, and application display. 2) The functions of the intelligent safety monitoring system of the Hekou Village reservoir are introduced for further enhancement. 3) Recommendations and tips are offered to develop safety surveillance systems for smart water conservation initiatives.	[13] (2020)
Water resources management	Water resources scheduling	1) A novel digital twin framework for Real-time monitoring and forecasting of water diversion projects; 2)A real-time self-correcting digital twin model for water systems offers enhanced monitoring, control, maintenance, and decision-making capabilities, improving system performance, efficiency, and resilience.	[16] (2023)
		1) A digital twin framework for optimizing the operational efficiency of Water Distribution Networks (WDN); 2) The digital twin in water distribution networks offers a comprehensive and data-driven approach to managing and optimizing the system, leading to improved operational efficiency, cost savings, and enhanced water quality management.	[17] (2023)
Equipment safety management	Hydraulic systems	1) A digital twin framework for fault diagnosis and predictive maintenance of hydraulic systems; 2) Using digital twin models in fault diagnosis and predictive maintenance for hydraulic systems offers significant advantages in accuracy, comprehensive information, cost reduction, and enhanced decision-making.	[15] (2022)
	Hydroelectric generating sets	1) A predictive maintenance model for electromechanical devices; 2) Provide the foundation for efficient data perception, storage, and analysis, leading to improved fault analysis, decision-making, and maintenance of water conservancy equipment.	[14] (2022)
	Large-scale hydro-steel structures	1) A digital twin-driven LS-HSS intelligent operation and maintenance platform is designed; 2) Key enabling technologies include real-time mapping, closed-loop control, dynamic health assessment, and intelligent decision-making; 3) The platform enables IoT connection, modeling and integration, virtual-real interaction, 2D/3D visualization, service expansion, etc., achieving a closed-loop proactive safety assurance and predictive maintenance mode with the stages of perception, analysis, decision-making, and optimization.	A DT-IOM platform for LS-HSS (This study)

updating, and functional integration (as depicted in Fig. 32). For instance, it integrates Unity's 3D visualization and interactive capabilities with Python's algorithm development and encapsulation functions, as well as ANSYS' multi-physics molding and simulation capabilities. Additionally, it fulfills the requirements for encapsulating LS-HSS intelligent operation and maintenance process data, algorithms, models, knowledge, and services while offering applicability, cost-effectiveness, and implementation efficiency advantages.

It has been successfully implemented in multiple hydraulic engineering projects such as Luhun Reservoir, Heiquan Reservoir, Hekoucun Reservoir, and Wuyue Pumped Storage Power Station. The following advantages are provided and validated, explicitly addressing the conventional operation and maintenance challenges.

(1) The high-fidelity real-time synchronization between the virtual and physical entity models of LS-HSS during operation has been achieved, enabling 2D/3D visualization monitoring and dynamic interaction under different conditions. Furthermore, a multi-scene, multi-terminal visualization tool has been developed to facilitate human–machine interaction in the 2D/3D virtual scene of LS-HSS operation. This enhances simulation verification for virtual inspections and device maintenance processes, remote operational guidance, and personnel virtual maintenance training capabilities.

(2) The DT-IOM platform can automatically detect and analyze security risks by integrating data, algorithms, knowledge, and services. Meanwhile, it enables real-time health status assessment of LS-HSS based on flood control scheduling or operation planning, combined with dynamic perception and inspection data.

(3) Considering factors such as actual water resource scheduling objectives, scheduling rules, and upstream and downstream flow conditions, the operational status within a foreseeable period can be dynamically simulated and calculated based on the DT-IOM platform. Moreover, intelligent decision-making solutions can be formulated by integrating online monitoring data, health assessment results, and simulation outcomes supported by a maintenance knowledge graph.

This study proposes a five-dimensional digital twin modeling framework for LS-HSS operation and maintenance, emphasizing the digital transformation pathway and platform of conventional operation and maintenance and the integration methods of data, models, knowledge, and services driven by digital twin technology. However, the actual operating conditions and demands constantly change, indicating unresolved matters, limitations, and deficiencies that require attention in future endeavors.

(1) Research on non-contact sensing technologies such as vision-based or acoustic-based sensors can further reduce qualitative evaluation indicators while enhancing digitalization in LS-HSS maintenance processes. Moreover, the accuracy and efficiency of a DT-IOM platform rely on adequate and effective data support. The data currently collected by the platform primarily consists of normal data, while insufficient and fragmented fault data may result in erroneous decision-making. How to effectively expand the fault data based on the DT-IOM platform and realize the accurate diagnosis and prediction of a small sample dataset requires further investigation.

(2) Further research is required to enhance precise modeling techniques that encompass geometric, physical, rule-based, and behavioral models within the virtual model to bridge the gap between theoretical models and practical applications (e.g., studying multi-field virtual model construction methods). In addition, the utilization of multi-layer neural networks in regression prediction reduces the necessity for sample points to be orthogonal, enabling the approximation of diverse nonlinear relationships through the design of distinct neural network structures and the selection of different activation functions. In future research, we will further explore optimization methods for LS-HSS surrogate models based on neural networks and conduct a more comprehensive analysis regarding the impact of basis functions on both predictive accuracy and generalization ability further to improve the surrogate model performance and computational efficiency.

(3) The integration of AI technology in data analysis, model optimization, and knowledge mining will further enhance the application value of the platform, so it is crucial to continuously improve its compatibility with emerging technologies and expand its functional capabilities. Furthermore, achieving a cost-effective balance between expenses and advantages is vital when constructing high-fidelity digital models with diverse characteristics and behaviors. Therefore, conducting a comprehensive study on the DT-IOM platform's maturity model and assessment techniques is imperative, incorporating the distinctive attributes and usage scenarios of the LS-HSS operation and maintenance process. This will facilitate the widespread adoption of the platform in modeling, integration, and service expansion.

Table 7
Representative research results of digital twin fundamental theory and implementation approach.

Research	Application fields	Theories or methods
Grieves et al. [30] (2015)	Fundamental modeling theory	A three-dimensional digital twin model
Alam et al. [33] (2017)		A Digital Twin Architecture Reference Model for the Cloud-Based Cyber-Physical Systems
Tao et al. [32] (2019)		A Five-dimensional digital twin model
Bönsch et al. [34] (2022)		A subject-oriented reference model for Digital Twins
Schroeder et al. [39] (2016)	Data exchange methods	Digital Twin Data Modeling with AutomationML and a Communication Methodology for Data Exchange
Redelinghuys et al. [35] (2020)		A six-layer architecture for the digital twin: a manufacturing case study implementation
Zhou et al. [41] (2021)	Modeling and updating	Digital-twin-driven geometric optimization of a centrifugal impeller with free-form blades for five-axis flank milling
Liu et al. [45] (2022)		Digital twin-enabled machining process modeling
Wang et al. [47] (2019)	Twin services	Digital Twin for rotating machinery fault diagnosis in smart manufacturing
Wang et al. [15] (2022)		Fault diagnosis and predictive maintenance for hydraulic system based on digital twin model

7. Conclusion and future works

This study proposes a five-dimensional digital twin modeling framework for LS-HSS operation and maintenance, specifically focusing on the digital transformation pathway of conventional operation and maintenance and the integration methods of data, models, knowledge,

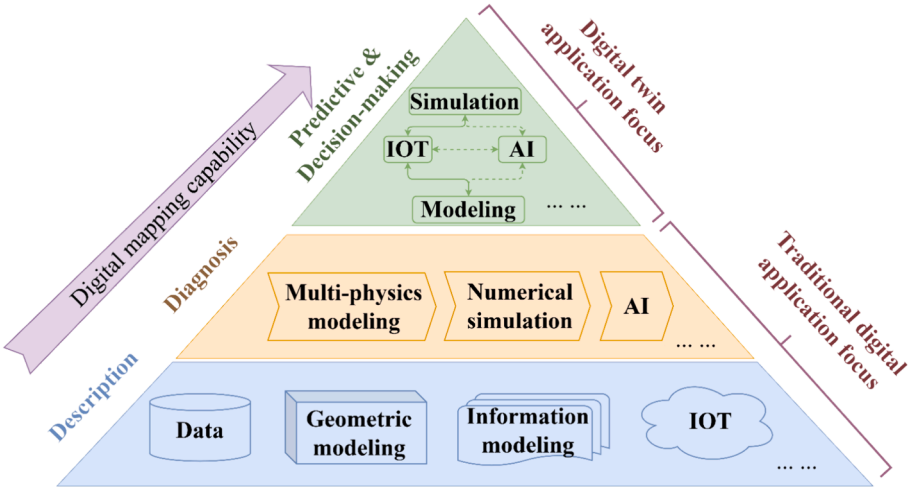


Fig. 31. An overview of the application focus of digital twin.

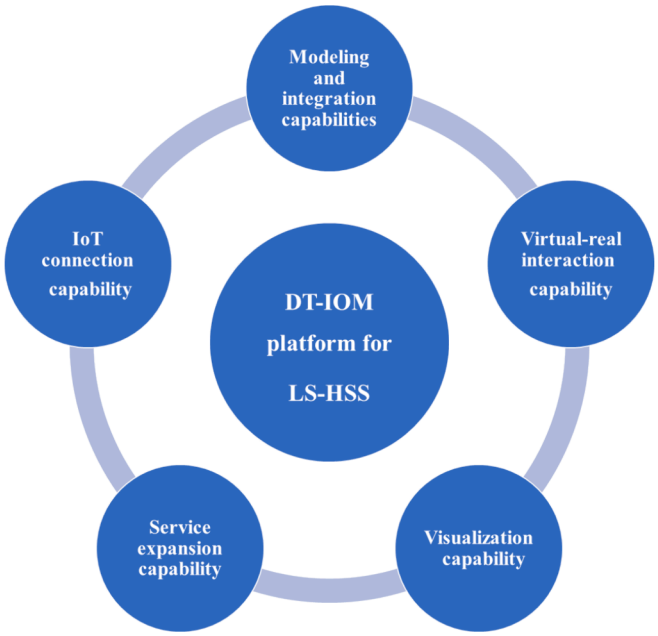


Fig. 32. The main functional advantages of the DT-IOM platform.

and services driven by digital twin technology. Secondly, key technologies and implementation methods for DT-IOM for LS-HSS are presented. It collects dynamic data on LS-HSS operations through sensors, embedded systems, data acquisition cards, or manual inspection systems and feeds it back to the virtual entity. Combining algorithms, models, and knowledge in the virtual entity for integrated analysis provides a decision-making basis for services and assists in optimizing LS-HSS operation and decision-making. Then, a comprehensive DT-IOM platform with real-time mapping, closed-loop control, dynamic health assessment, and intelligent decision-making is constructed and applied to establish a closed-loop proactive safety assurance and predictive maintenance mode with the stages of perception, analysis, decision-making, and optimization. The DT-IOM platform consisted of a physical entity layer, a digital twin layer, a connection layer, and a service layer. The primary data on the physical entity are acquired and processed for 2D/3D visualization display, process control, and decision optimization. The digital twin platform provides the data, model, and knowledge platform services, enabling the integration of virtual and physical components. The operation and maintenance knowledge databases based on the knowledge graph are established to obtain the structured or unstructured and semi-structured knowledge by analyzing the simulation databases, rule files, standard files, and maintenance manuals of LS-HSS. The high-fidelity 3D virtual representation of the actual LS-HSS is created using 3ds Max. Finally, a comprehensive analysis is conducted on the functionality and advantages of the DT-IOM platform for LS-HSS. Compared to the existing fundamental theories and platforms, which exhibit limitations and deficiencies in terms of applicability, cost, and implementation efficiency, the DT-IOM platform for LS-HSS based on practical engineering applications highlights the following functional advantages: IoT connectivity capability, modeling & integration capabilities, virtual-real interaction capability, 2D/3D visualization capability, and service expansion capability.

The main conclusions are as follows:

(1) The proposed five-dimensional digital twin modeling framework for LS-HSS operation and maintenance seamlessly integrates five levels of physical entity, virtual entity, digital twin data, services, and connections, providing a practical digital transformation pathway. This includes integrating data, models, knowledge, and services driven by digital twin technology to establish a closed-loop proactive safety assurance and predictive maintenance mode with the stages of perception, analysis, decision-making, and optimization.

Table 8
Comparison of commonly used digital twin platforms.

Categories	Characteristics	Application object/ level
ANSYS–Twin Builder	1) Support for multiple multi-physics field model reduction methods and modeling languages 2) Multi-physics digital twin of equipment	Component/ equipment level
ABB–PickMaster Twin	1) Built-in digital twins supporting virtual debugging 2) Testing robot configuration based on virtual assembly line	Equipment/ production line level
SIEMENS–PlantSight	1) Integrating factory construction, operation information, and digital models 2) Build a digital factory to dynamically analyze the operational status of the factory and equipment health	Production line/ factory level
Unity–Unity Pro	1) It involves the entire process of modeling, behavior description, and human–computer interaction 2) Real-time rendering, 3D visualization, dynamic integration	Factory/ industrial park/ city-level
Dassault–3DEXPERIENCE	1) Integrate all Dassault digital tools into the platform 2) Applicable to situations where there are twin requirements for the device itself and the device running environment	Factory/ industrial park/ city-level
Microsoft–Azure Digital Twins	1) Easily simulate and create digital representations of connected environments with an open modeling language 2) Bring digital twins to life with real-time graphical representation	Factory/ industrial park/ city-level
DT-IOM platform for LS-HSS (This study)	1) Capability of IoT connectivity, modeling and integration, virtual-real interaction, 2D/3D visualization, and service expansion for LS-HSS 2) Fulfill the diverse requirements for encapsulating data, algorithms, models, knowledge, and services of LS-HSS intelligent operation and maintenance process 3) Possess advantages in terms of applicability, cost-effectiveness, and implementation efficiency	Hydraulic engineering Component/ equipment level

(2) The DT-IOM platform, equipped with a real-time 2D/3D visualization feature, online monitoring capability, dynamic health assessment functionality, and intelligent decision-making capabilities, offers advantages in providing a differentiated functional configuration for various levels of digitization. It facilitates dynamic perception, evaluation, and decision optimization of the operational status of LS-HSS and significantly enhances the capacity to operate safely and reliably under intricate working conditions. The on-site data demonstrates that LS-HSS operation and maintenance, based on DT-IOM, effectively improves the digital recognition capabilities of performance evolution characteristics for LS-HSS during long-term service. The average equipment inspection time is reduced by approximately 50 %, while operation and maintenance response and processing efficiency are improved by over 60 %. Moreover, the equipment failure rate decreases by over 20 %. China's

Ministry of Water Resources Science and Technology Promotion Center has incorporated The DT-IOM platform into the Key Promotion and Guidance Catalogue for Advanced Practical Technologies (TZ2021277). It will continue to drive forward the digital and intelligent transformation of LS-HSS operation and maintenance processes in the future.

(3) The accuracy and efficiency of a DT-IOM platform rely on adequate and effective data support. The presence of inadequate and fragmented fault data may lead to erroneous decision-making. How to effectively expand the fault data and realize the accurate diagnosis and prediction of a small sample dataset is a problem that needs further study. In addition, achieving a cost-effective balance between expenses and advantages is crucial when constructing high-fidelity digital models for complex equipment with diverse characteristics and behaviors. Therefore, it is essential to study the maturity model and assessment techniques for DT-IOM by incorporating the unique attributes and usage scenarios of the LS-HSS operation and maintenance process. In the future, integrating AI technology in data analysis, model optimization, and knowledge mining will further enhance the application value of the platform, leveraging its compatibility with emerging technologies and expanding functional capabilities. Consequently, this will elevate model updates, algorithm optimization, knowledge integration, and service encapsulation on the platform.

CRediT authorship contribution statement

Helin Li: Writing – original draft, Visualization, Validation, Software, Methodology, Data curation. **Rui Zhang:** Writing – review & editing, Supervision, Conceptualization. **Shufeng Zheng:** Validation, Software, Investigation. **Yonghao Shen:** Visualization, Validation, Software. **Chunjian Fu:** Validation, Software, Data curation. **Huadong Zhao:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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